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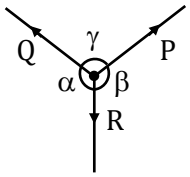
Basic Mathematics Used in Physics and Vectors

1. If three unit vectors are inclined at an angle of 60° with each other, then the magnitude of their resultant vector will be :-
(1) zero (2) $\sqrt{6}$ (3) $\sqrt{3}$ (4) $\sqrt{9}$
PBM250

2. A vector is represented by $3\hat{i}+\hat{j}+2\hat{k}$. Its length in XY plane is :-
(1) 2 (2) $\sqrt{14}$ (3) $\sqrt{10}$ (4) $\sqrt{5}$
PBM251

3. The unit vector parallel to the resultant of the vectors $\vec{A}=4\hat{i}+3\hat{j}+6\hat{k}$ and $\vec{B}=-\hat{i}+3\hat{j}-8\hat{k}$ is :-
(1) $\frac{1}{7}(3\hat{i}+6\hat{j}-2\hat{k})$ (2) $\frac{1}{7}(3\hat{i}+6\hat{j}+2\hat{k})$
(3) $\frac{1}{49}(3\hat{i}+6\hat{j}-2\hat{k})$ (4) $\frac{1}{49}(3\hat{i}-6\hat{j}+2\hat{k})$
PBM252

4. A body is in equilibrium under the action of three coplanar forces P, Q and R as shown in the figure. Select the correct statement :-



(1) $\frac{P}{\sin\alpha}=\frac{Q}{\sin\beta}=\frac{R}{\sin\gamma}$
(2) $\frac{P}{\cos\alpha}=\frac{Q}{\cos\beta}=\frac{R}{\cos\gamma}$
(3) $\frac{P}{\tan\alpha}=\frac{Q}{\tan\beta}=\frac{R}{\tan\gamma}$
(4) $\frac{P}{\sin\beta}=\frac{Q}{\sin\gamma}=\frac{R}{\sin\alpha}$
PBM253

5. The algebraic sum of two co-initial vectors is 16 units. Their vector sum is 8 units and the resultant of the vectors is perpendicular to the smaller vector. Then the magnitudes of the two vectors are-
(1) 2 unit and 14 unit
(2) 4 unit and 12 unit
(3) 6 unit and 10 unit
(4) 8 unit and 8 unit
PBM254

6. A truck travelling due north at 20 m/s turns west and travels at the same speed. The change in its velocity is :-
(1) 40 m/s N – W
(2) $20\sqrt{2}$ m/s N – W
(3) 40 m/s S – W
(4) $20\sqrt{2}$ m/s S – W
PBM255

7. A metal ring is being heated so that at any instant of time t in second, its area is given by

$$A=\left(3t^2+\frac{t}{3}+2\right)m^2$$

What will be the rate of increase of area at t = 10 sec.
(1) $\frac{185}{3}\frac{m^2}{sec}$ (2) $\frac{181}{3}\frac{m^2}{sec}$
(3) $\frac{181}{5}\frac{m^2}{sec}$ (4) $\frac{181}{7}\frac{m^2}{sec}$
PBM256

8. The angle between vectors $\vec{A}$ and $\vec{B}$ is 60°. What is the ratio of $\vec{A}\cdot\vec{B}$ and $|\vec{A}\times\vec{B}|$?
(1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{1}{\sqrt{4}}$ (4) $\frac{1}{\sqrt{5}}$
PBM257

9. The magnitude of x and y components of $\vec{A}$ are 7 and 6 respectively. Also, the magnitudes of x and y components of $\vec{A}+\vec{B}$ are 11 and 9 respectively. Calculate the magnitude of vector $\vec{B}$.
(1) 10 (2) 5 (3) 6 (4) 3
PBM258

10. Find the sum of series

$$2q+\frac{q}{3}+\frac{q}{9}+\frac{q}{27}+.....$$

(1) $\frac{3}{2}q$ (2) $\frac{7}{2}q$ (3) $\frac{5}{2}q$ (4) $\frac{5}{9}q$
PBM259

11. If the sum of two unit vectors is a unit vector, then magnitude of their difference is :-
(1) $\sqrt{2}$ (2) $\sqrt{3}$
(3) $1/\sqrt{2}$ (4) $\sqrt{5}$
PBM260

12. Two vectors $\vec{A}$ and $\vec{B}$ lie in a plane, another vector $\vec{C}$ lies outside this plane, then the resultant of these three vectors is $\vec{A}+\vec{B}+\vec{C}$.
(1) Can be zero
(2) Cannot be zero
(3) Lies in the plane containing $\vec{A}+\vec{B}$
(4) Lies in the plane containing $\vec{C}$
PBM261

13. Find the angle of tangent drawn to the curve $y = 3x^2 - 7x + 5$ at the point (1,1) with the positive x-axis :-
(1) $\theta = 60°$ (2) $\theta = 120°$
(3) $\theta = 135°$ (4) $\theta = 45°$
PBM262

14. Let the angle between two non-zero vectors $\vec{A}$ and $\vec{B}$ be 120° and their resultant is $\vec{C}$, then :-
(1) $|\vec{C}|$ must be equal to $|\vec{A}-\vec{B}|$
(2) $|\vec{C}|$ must be less than $|\vec{A}-\vec{B}|$
(3) $|\vec{C}|$ must be greater than $|\vec{A}-\vec{B}|$
(4) $|\vec{C}|$ may be equal to $|\vec{A}-\vec{B}|$
PBM263

15. Two forces 3N and 2N are at an angle θ such that the resultant is R. The first force is now increased to 6N and the resultant become 2R. The value of θ is: -
(1) 30° (2) 60° (3) 90° (4) 120°
PBM264

16. If two vectors $2\hat{i}+3\hat{j}-\hat{k}$ and $-4\hat{i}-6j+\lambda\hat{k}$ are parallel to each other then value of λ be :-
(1) 0 (2) 2 (3) 3 (4) 4
PBM265

17. The value of $\cos\alpha = \frac{12}{13}$ then the value of sin 2α will be :-
(1) $\frac{120}{169}$ (2) $\frac{169}{120}$ (3) $\frac{159}{120}$ (4) $\frac{223}{240}$
PBM266

18. If $\vec{A}\times\vec{B}=\vec{C}$, then which of the following statement is wrong :-
(1) $\vec{C}\perp\vec{A}$ (2) $\vec{C}\perp\vec{B}$
(3) $\vec{C}\perp(\vec{A}+\vec{B})$ (4) $\vec{C}\perp(\vec{A}\times\vec{B})$
PBM267

19. The value of area bounded by f(y) = sinx from $x = \pi$ to $x = 2\pi$ is :-
(1) 1 (2) –1 (3) 2 (4) –2
PBM268

20. The vector $\vec{P}=a\hat{i}+a\hat{j}+3\hat{k}$ and $\vec{Q}=a\hat{i}-2\hat{j}-\hat{k}$ are perpendicular to each other. The positive value of a is :-
(1) 3 (2) 4 (3) 9 (4) 13
PBM269

21. A vector $\vec{F}_1$ is along the positive X-axis. If its vector product with another vector $\vec{F}_2$ is zero, then $\vec{F}_2$ could be :-
(1) $4\hat{j}$ (2) $-(\hat{i}+\hat{j})$
(3) $(\hat{j}+\hat{k})$ (4) $(-4\hat{i})$
PBM270

22. What is the unit vector perpendicular to the following vectors $2\hat{i}+2\hat{j}-\hat{k}$ and $6\hat{i}-3\hat{j}+2\hat{k}$:
(1) $\frac{\hat{i}+10\hat{j}-18\hat{k}}{5\sqrt{17}}$ (2) $\frac{\hat{i}-10\hat{j}+18\hat{k}}{5\sqrt{17}}$
(3) $\frac{\hat{i}-10\hat{j}-18\hat{k}}{5\sqrt{17}}$ (4) $\frac{\hat{i}+10\hat{j}+18\hat{k}}{5\sqrt{17}}$
PBM271

23. What is the value of linear velocity, if $\vec{\omega}=3\hat{i}-4\hat{j}+\hat{k}$ and $\vec{r}=5\hat{i}-6\hat{j}+6\hat{k}$
(1) $6\hat{i}-2\hat{j}+3\hat{k}$ (2) $6\hat{i}-2\hat{j}+8\hat{k}$
(3) $4\hat{i}-13\hat{j}+6\hat{k}$ (4) $-18\hat{i}-13\hat{j}+2\hat{k}$
PBM272

24. If a vector $\vec{P}$ making angles α, β, and γ respectively with the X, Y and Z axes respectively. Then $\sin^2\alpha + \sin^2\beta + \sin^2\gamma$ =
(1) 0 (2) 1 (3) 2 (4) 3
PBM273

25. The vectors from origin 'O' to the points A and B are $\vec{A}=3\hat{i}-6j+2\hat{k}$ and $\vec{B}=2\hat{i}+j-2\hat{k}$ respectively. The area of the triangle OAB be :-
(1) $\frac{5}{2}\sqrt{17}$ (2) $\frac{2}{5}\sqrt{17}$
(3) $\frac{3}{5}\sqrt{17}$ (4) $\frac{5}{3}\sqrt{17}$
PBM274

26. If the direction cosines of a vector along y-axis and z-axis is $\frac{2}{\sqrt{14}}$ and $\frac{3}{\sqrt{14}}$ respectively and the vector having magnitude $2\sqrt{14}$. Then the vector is -
(1) $\hat{i}+\hat{j}+2\hat{k}$ (2) $-2\hat{i}+\hat{j}-\hat{k}$
(3) $6\hat{i}+2\hat{j}+4\hat{k}$ (4) $2\hat{i}+4\hat{j}+6\hat{j}$
PBM275

27. Given below are two statements : One is labelled as 'Assertion (A)' and 'Reason (R)'.
Assertion (A): Two vectors in three dimension space are always coplanar.
Reason (R): For the coplanarity of vector in space, then their scalar triple product must be zero.

$$\left(\vec{A}\times\vec{B}\right)\cdot\vec{C}=0$$

In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is correct explanation of (A)
(3) (A) is false but (R) is true.
(4) (A) is true but (R) is false.

PBM276

28. Differentiate sin (cos (x^2)) with respect to x
(1) $\cos^2(x^2)$
(2) cos (sin(x^2)
(3) $\sin^2(x^2)$
(4) $-2x\cos(\cos(x^2))\sin x^2$

PBM277

29. Integration of x^n i.e. $\int x^n dx=\frac{x^n}{n+1}+C$ is valid for the value of n?
(1) all number lying on real axis
(2) only positive real numbers
(3) only negative real numbers
(4) All real numbers except n = –1

PBM278

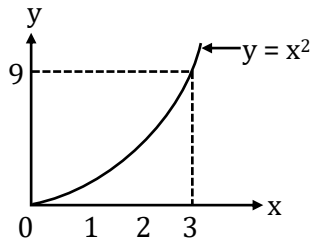
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(1) $4y = 7x + 9, c = \frac{9}{5}$
(2) $3y = 4x - 10, c = -\frac{10}{3}$
(3) $3x - 4y + 10 = 0, c = \frac{5}{2}$
(4) $y = 4x + 5, c = 5$

PBM279

31. What can be the possible values of angle between two vectors $\vec{A}$ and $\vec{B}$
(a) 60° (b) 210° (c) –240° (d) 120°
(1) only (a)
(2) only (a), (b) and (d)
(3) (a) and (d)
(4) all values are possible

PBM280

32. Calculate the area enclosed between the curve $y = x^2$ and y-axis



(1) $\frac{19}{3}$ (2) 9 (3) $\frac{17}{3}$ (4) 18

PBM281

33. The sum of the given series is
$1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+\frac{1}{16}+\ldots\ldots\ldots\ldots$
(1) $\frac{16}{25}$ (2) $\frac{5}{9}$ (3) $\frac{2}{3}$ (4) $\frac{8}{3}$

PBM282

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	1	1	3	4	2	2	2	3	2	2	3	2	4
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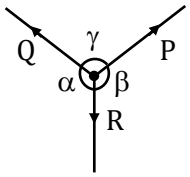
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PBM263

15. Two forces 3N and 2N are at an angle θ such that the resultant is R. The first force is now increased to 6N and the resultant become 2R. The value of θ is: -
(1) 30° (2) 60° (3) 90° (4) 120°
PBM264

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PBM267

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PBM268

20. The vector $\vec{P}=a\hat{i}+a\hat{j}+3\hat{k}$ and $\vec{Q}=a\hat{i}-2\hat{j}-\hat{k}$ are perpendicular to each other. The positive value of a is :-
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PBM270

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23. What is the value of linear velocity, if $\vec{\omega}=3\hat{i}-4\hat{j}+\hat{k}$ and $\vec{r}=5\hat{i}-6\hat{j}+6\hat{k}$
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PBM274

26. If the direction cosines of a vector along y-axis and z-axis is $\frac{2}{\sqrt{14}}$ and $\frac{3}{\sqrt{14}}$ respectively and the vector having magnitude $2\sqrt{14}$. Then the vector is -
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PBM276

28. Differentiate sin (cos (x^2)) with respect to x
(1) $\cos^2(x^2)$
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(3) $\sin^2(x^2)$
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PBM278

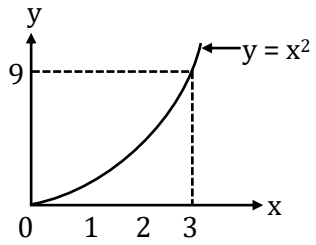
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(1) $\frac{19}{3}$ (2) 9 (3) $\frac{17}{3}$ (4) 18

PBM281

33. The sum of the given series is
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(1) $\frac{16}{25}$ (2) $\frac{5}{9}$ (3) $\frac{2}{3}$ (4) $\frac{8}{3}$

PBM282

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	1	1	3	4	2	2	2	3	2	2	3	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
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Question	31	32	33												
Answer	3	4	3												

Units, Dimensions and Measurement

1. How many number of base units are contained in S.I. system of units?
(1) Three (2) Five (3) Nine (4) Seven
PPM217

2. In the following given options, which of the two base units have not been included in the S.I. system.
(1) mole, candela (2) radian, steradian
(3) kelvin, ampere (4) kelvin candela
PPM218

3. Which of the following quantities have units for measurement but are dimensionless in nature?
(1) Coefficient of friction (μ), efficiency of heat engine (η)
(2) Dielectric constant (k), permittivity of free space ($\in_0$)
(3) Plane angle (θ), solid angle (Ω)
(4) Relative permeability (μ_r), magnetic susceptibility (χ)
PPM219

4. From the given physical quantities which one is derived quantity according to S.I. system
(1) Luminous intensity (2) Charge
(3) Time (4) Electric current
PPM220

5. A Vernier calliper can measure the length and thickness of object in the minimum range of ..... m.
(1) 10^{-1} (2) 10^{-2} (3) 10^{-4} (4) 10^{-3}
PPM221

6. Screw gauge and spherometer can be used to measure length as small as ..... m.
(1) 10^{-3} (2) 10^{-2} (3) 10^{2} (4) 10^{-5}
PPM222

7. Which method is used for measuring large distances such as the distance of planets or a star from the earth?
(1) odometer
(2) optical range finder
(3) Parallax method
(4) Laser distance measurers.
PPM223

8. A metre scale is used for measuring length from?
(1) 1 cm to 100 cm
(2) 0.1 cm to 1000 cm
(3) 1m to 100m
(4) 10^{-3} m to 10^{2}m
PPM224

9. Given below are two statements: - One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): - When two quantities are added or subtracted, the absolute error in the final result is the sum of absolute error in the individual quantities.
Reason (R): - When two quantities are multiplied or divided, the absolute error in the result is the sum of the absolute errors in the multipliers in the light of the above statements, choose the most appropriate answers from the options given below: -
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true.
PPM225

10. How many significant figures are there in given numbers 0.002308, 12300, 3.500 and 0.06900 respectively.
(1) 6,5,2,2 (2) 7,5,4,6
(3) 4,3,4,4 (4) 5,3,2,2
PPM226

11. Given below are two statement; one is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion (A): - In multiplication or division, the final result should retain as many significant figures as are there in the original number with least significant figures.
Reason (R): - In addition or subtraction, the final result should retain as many decimal places as are there in the numbers with the least decimal places.
(1) Both (A) and (R) are true and (R) is correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true
PPM227

12. In column-I, some of the physical quantities are listed and column-II includes the dimensional formulas for the given quantities.

Match the following

Column-I		Column-II	
(A)	Wein constant	(P)	$[ML^3T^{-3}A^{-1}]$
(B)	Electric flux	(Q)	$[M^0L^{-1}T^0]$
(C)	Wave numbers	(R)	$[M^0LT^0K]$
(D)	Magnetic flux density	(S)	$[ML^0T^{-2}A^{-1}]$

(1) (A)-(P), (B)-(Q), (C)-(R),(D)-(S)
(2) (A)-(R), (B)-(P), (C)-(Q),(D)-(S)
(3) (A)-(S), (B)-(Q), (C)-(P),(D)-(R)
(4) (A)-(R), (B)-(Q), (C)-(P),(D)-(S)

PPM228

13. In column-I, various physical quantities are list and column-II include the dimensional for the given quantities match the following.

Column-I		Column-II	
(A)	Surface tension	(P)	$[M^0L^0T^0]$
(B)	Gravitational constant	(Q)	$[M^1L^2T^0]$
(C)	Angular displacement	(R)	$[M^{-1}L^3T^2]$
(D)	Moment of inertia	(S)	$[M^1L^0T^{-2}]$

(1) (A)-(S), (B)-(R), (C)-(P),(D)-(Q)
(2) (A)-(R), (B)-(S), (C)-(Q),(D)-(P)
(3) (A)-(Q), (B)-(P), (C)-(R),(D)-(S)
(4) (A)-(S), (B)-(P), (C)-(R),(D)-(Q)

PPM229

14. If we have two quantities A and BC^2 such that the dimensional formula of A is $[M^0LT^{-2}]$ and for B is $[M^0L^1T^0]$. Then which of the following case is possible

(1) Given quantities can be added as well as multiplied irrespective of dimension of C
(2) Given quantities can be added only if dimension of C is $[T^{-1}]$ and multiplied for any dimension of C
(3) Given quantities can be multiplied if dimension of C is $[T^{-1}]$ and can be added for any dimension of C
(4) Given quantities can be added as well as multiplied if only if dimension of C is $[T^{-1}]$

PPM230

15. Fill in the blank

$G = 6.67 \times 10^{-11} Nm^2 (kg)^{-2} = \ldots\ldots\ldots (cm)^3 s^2 g^{-1}$

(1) 6.67×10^{-12} (2) 6.67×10^{-14}
(3) 6.67×10^{-8} (4) 6.67×10^{-9}

PPM231

16. The physical quantity that has the same dimensional formula as acceleration is: -

(1) Electric field intensity
(2) Gravitation field intensity
(3) Energy density
(4) Self inductance

PPM232

17. If I represents the intensity of EM wave and c represent speed of light then $\frac{I}{c}$ has the same dimension as

(1) Resistance (2) Momentum
(3) Pressure (4) Velocity

PPM233

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	3	2	3	4	3	4	3	3	2	2	1	2	3
Question	16	17													
Answer	2	3													

Kinematics

1. The location of a particle is changed. What can we say about the displacement & distance covered by the particle?
(1) Both cannot be zero
(2) One of the two may be zero
(3) Both must be zero
(4) Both must be equal

PMP318

2. A particle is moving in influence of constant acceleration. It's path may be :
(1) Circle (2) Hyperbola
(3) Parabola (4) Ellipse

PMP319

3. Which of the following statement(s) is/are wrong :
(A) A body can have constant speed & varying acceleration
(B) A body can have zero velocity & uniform acceleration
(C) A body can have constant velocity & uniform acceleration
(1) Only C (2) Both B & C
(3) Both A & C (4) All A, B, C

PMP320

4. For a moving object, the numerical value of the ratio of magnitude of instantaneous velocity to instantaneous speed is :
(1) Always less than 1
(2) Always equal to 1
(3) Always more than 1
(4) Equal to or less than one

PMP321

5. Two bodies of masses 1kg & 10 kg are dropped simultaneously from the top of a tower. On the surface of earth :
(1) Heavier will reach first
(2) Lighter will reach first
(3) Both will reach simultaneously
(4) Any one will reach first

PMP322

6. A body is thrown upward & reaches it's maximum height. At this position :
(1) It's velocity is zero & it's acceleration is also zero
(2) It's velocity is zero but it's acceleration is maximum
(3) It's acceleration is minimum
(4) It's velocity is zero & it's acceleration is non zero due to gravity

PMP323

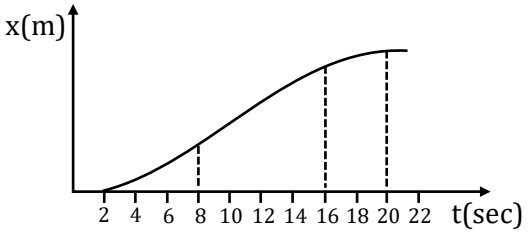
7. Mark the correct statement :
(1) If velocity & acceleration have opposite sign, then speed of particle will increase
(2) If velocity is zero at a instant, then acceleration of particle will be zero at that instant
(3) If velocity is zero in a time interval then acceleration is not equal to zero at any instant within the time interval
(4) A particle moves along x-axis. If it's x-coordinate & velocity have opposite sign then the particle will move towards origin

PMP324

8. Which of the following statements is wrong about a ball thrown vertically up :
(A) It is moving with constant acceleration
(B) It may have different velocities at the same position
(C) It may have two positions at the same time
(D) The angular momentum of particle about point of projection remain conserved
(1) only C (2) C, D
(3) B, C, D (4) A, B, C, D

PMP325

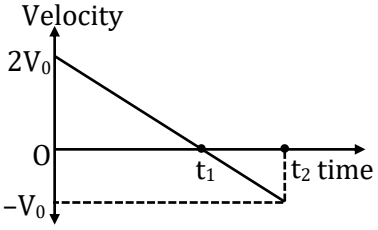
9. What is nature of acceleration between t = 18 s to t = 20 s in below diagram



(1) Positive (2) Negative
(3) Zero (4) None of these

PMP326

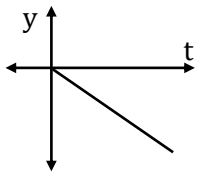
10. In the given graph which of following physical quantities changes their sign?

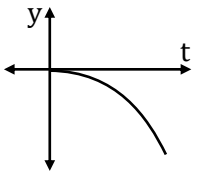


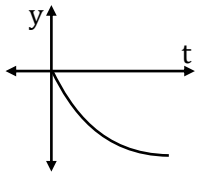
(1) Displacement (2) Velocity
(3) Both (1) and (2) (4) Acceleration

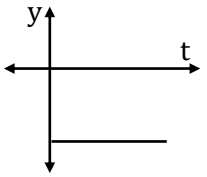
PMP327

11. Which of following graph represents displacement-time of an object under free fall. Neglect air resistance.

(1) 

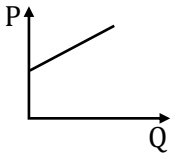
(2) 

(3) 

(4) 

PMP328

12. Variation of quantity P with quantity Q is plotted in given figure, describes motion of particle in a straight line. Then which of the following statement is wrong :



(1) Quantity Q may represent time
(2) Quantity P is velocity if motion is uniform
(3) Quantity P may be displacement if motion is uniform
(4) Quantity P may be velocity if motion is uniformly accelerated

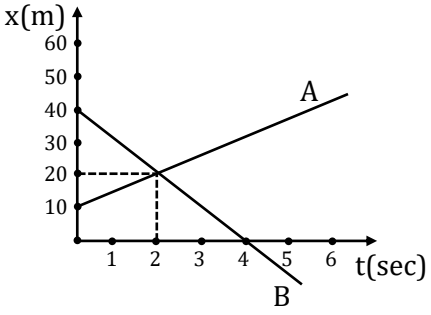
PMP329

13. A ball is dropped from a building of height 45m. Simultaneously another ball is thrown up with a speed of 40 m/s. Calculate the magnitude of relative velocity of the balls when both balls are in air :

(1) 0 m/s (2) 20 m/s
(3) 40 m/s (4) 60 m/s

PMP330

14. Displacement-time curve of two objects A & B is shown in the figure. Find relative velocity of B with respect to A.



(1) – 15 m/s (2) – 10 m/s
(3) 15 m/s (4) 10 m/s

PMP331

15. A lift coming from 8th floor is just about to reach 4th floor and stops on 4th floor. Taking ground as origin and taking upward direction as positive for all quantities, which of following is correct?

(1) $x < 0, v < 0, a < 0$ (2) $x > 0, v < 0, a < 0$
(3) $x > 0, v < 0, a > 0$ (4) $x > 0, v > 0, a < 0$

PMP332

16. A ball A is dropped from a building of height 45 m. Simultaneously another identical ball B is thrown up with a speed 50 ms^{-1}. The relative speed of ball B w.r.t. ball A at any instant of time when both the balls will be in air (Take $g = 10\ ms^{-2}$)

(1) 0 (2) 10 ms^{-1}
(3) 25 ms^{-1} (4) 50 ms^{-1}

PMP333

17. Two objects are moving on straight line with non-zero velocities in opposite directions, then magnitude of relative velocity will be

(1) Equal to individual velocity
(2) Less than individual velocities
(3) Greater than magnitude of individual velocities
(4) Greater or equal to magnitude of individual velocities.

PMP334

18. Two particles A and B are moving on a straight line with the same speed. Which of the following statement is/are correct for the relative motion of the two particles :

(1) The relative velocity v_{AB} or v_{BA} is zero only if they are moving in the same direction
(2) If the particles are moving in opposite direction, the magnitude of v_{BA} or v_{AB} is twice than the magnitude of velocity of A or that of B.
(3) The relative velocity v_{AB} or v_{BA} is always zero
(4) Both (1) and (2)

PMP335

19. The relation $\langle \vec{v} \rangle = \frac{\vec{u}+\vec{v}}{2}$ is valid for

(1) Constant acceleration only
(2) Non-uniform acceleration only
(3) Uniform motion only
(4) Uniform and non-uniform acceleration both

PMP336

20. Which of the following statement is/are Incorrect?
(I) For motion involving change in direction along the straight line, displacement may decrease.
(II) Direction of acceleration is same as direction of velocity.
(III) Average speed is, in general greater than or equal to the magnitude of the average velocity.
(IV) Equation of motion is valid only in one dimensional uniform accelerated motion
(1) only I (2) only IV
(3) II, IV (4) I, III, IV

PMP337

21. A person standing on the ground throws a stone upwards, then the path of the stone as seen by the person in a train moving on a straight horizontal track with constant speed-
(1) Circular (2) Parabolic
(3) Straight line (4) Ellipse

PMP338

22. A boy tries to cross the river and want to reach at the exactly opposite point on the other bank then what is the necessary relation between v_b and v_r :-
$v_b \rightarrow$ velocity of boy in still water ;
$v_r \rightarrow$ velocity of river
(1) $v_b < v_r$ (2) $v_b = v_r$
(3) $v_b > v_r$ (4) None of these

PMP339

23. A marble 'A' is dropped vertically. Another identical marble 'B' is projected horizontally from the same point at the same instant then-
(If friction of air considered as negligible)
(1) A will reach the ground earlier than B
(2) B will reach the ground earlier than A
(3) Both A and B will reach the ground at the same instant
(4) None of these

PMP340

24. Which of the following statement is correct -
(1) Direction of velocity at any point on path is tangential to the path
(2) Direction of average velocity is along direction of motion during that interval
(3) Direction of motion is always along direction of acceleration
(4) A particle may undergo 2-dimensional motion even if acceleration & velocity remain in same direction

PMP341

25. In which of the following examples average speed is greater than the magnitude of average velocity.
(1) Projectile motion (2) Circular motion
(3) Planetary motion (4) All of the above

PMP342

26. For projectile motion
(1) Acceleration must be constant
(2) Initial momentum and force must be noncollinear
(3) Velocity, must vary with time
(4) All of the above

PMP343

27. If $\vec{v}$ and $\vec{a}$ represent velocity and acceleration respectively, then $\hat{v} = \hat{a}$ may remains applicable in -
(1) 1-Dimensional motion
(2) Projectile motion
(3) Circular motion
(4) None of the above

PMP344

28. Which of the following sets of factors will affect the horizontal distance covered by an athlete in a long-jump event.
(1) Speed before he jumps and his weight
(2) The direction in which he leaps and the initial speed
(3) The force with which he pushes the ground and his speed
(4) None of these

PMP345

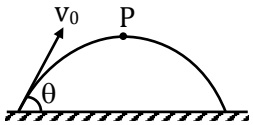
29. A ball is projected at some angle from surface, when it return to surface then it's change in momentum will be :-
(1) acceleration due to gravity × time of flight
(2) weight of ball × half of time of flight
(3) weight of ball × time of flight
(4) weight of ball × range

PMP346

30. A projectile is projected with speed u at an angle 2θ from horizontal, speed of projectile at the instant when it subtend angle θ from horizontal is :-
(1) $\frac{u\cos 2\theta}{2}$
(2) u (2 cos θ – sec θ)
(3) u cos θ
(4) u (cos θ – sec θ)

PMP347

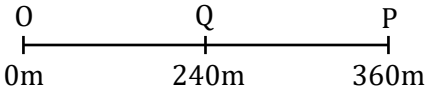
31. In a ground to ground projectile motion tangential acceleration at the topmost point P of the trajectory is :-



(1) g (2) gcosθ
(3) 0 (4) None of these

PMP348

32. A car is moving along a straight line OP as shown in the figure. It moves from O to P in 18 s and returns from P to Q in 6s and Q to O in 10 s. Which of the following statement is correct regarding the motion of the car?



(1) The average speed of the car in going from O to P and come back to Q is 20 ms^{-1}.
(2) The average velocity of the car in going from O to P and come back to Q is 20 ms^{-1}.
(3) The average speed of the car in going from O to P and come back to O is 20 ms^{-1}.
(4) The average velocity of the car in going from O to P and come back to O is 20 ms^{-1}.

PMP349

33. A body moving along a straight line travels one fourth of the total distance with a speed of 3.0 ms^{-1}. The remaining distance is covered with a speed of 4.0 ms^{-1} for half the time and 5.0 ms^{-1} for the other half of the time. The average speed during the motion is :

(1) 4.0 ms^{-1} (2) 6.0 ms^{-1}
(3) 3.8 ms^{-1} (4) 2.4 ms^{-1}

PMP350

34. A boy walks on a straight road from his home to a market 2.5 km with a speed of 5 km h^{-1}. Finding the market closed he instantly turns and walks back with a speed of 7.5 km h^{-1}. What is the average speed and average velocity of the boy between t = 0 to t = 50 min?

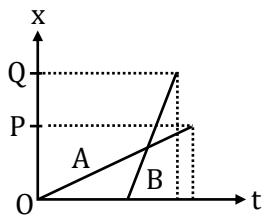
(1) 0, 0 (2) 6 km h^{-1}, 0
(3) 0, 6 km h^{-1} (4) 6 km h^{-1}, 6 km h^{-1}

PMP351

35. Speedometer of a car measures
(1) Average speed
(2) Average velocity
(3) Instantaneous speed
(4) Instantaneous velocity

PMP352

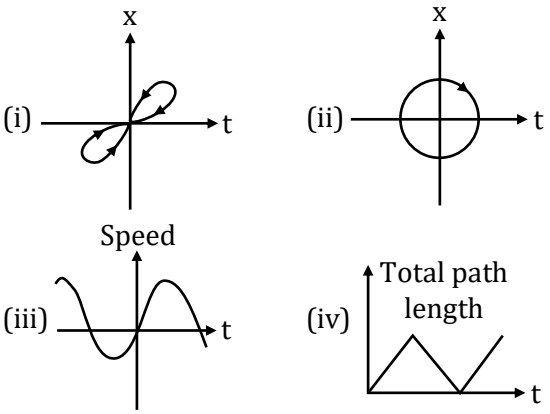
36. The position-time (x-t) graphs for two children A and B returning from their school O to their homes P and Q respectively are as shown in the figure. Choose the incorrect statement regarding these graphs.



(1) A lives closer to the school than B.
(2) A starts from the school earlier than B.
(3) A walks faster than B.
(4) A and B reach home at different time.

PMP353

37. Which of the following graphs cannot possibly represent one dimensional motion of a particle?



(1) (i) and (ii) (2) (ii) and (iii)
(3) (i), (ii) and (iii) (4) All four

PMP354

38. A body sliding on a smooth inclined plane requires 4 seconds to reach the bottom, starting from rest at the top. How much time does it take to cover one-fourth the distance starting from rest at the top?

(1) 1 s (2) 4 s (3) 2 s (4) 16 s

PMP355

39. The displacement time graph of a moving object is a straight line. Then,
(1) Its acceleration may be uniform
(2) lts velocity may be uniform
(3) Its acceleration may be variable
(4) Both its velocity and acceleration may be uniform

PMP356

40. If the displacement of an object is zero, then what can we say about its distance covered?
(1) It is negative
(2) It is must be zero
(3) It cannot be zero
(4) It may or may not be zero

PMP357

41. Which of the following changes when a particle is moving with uniform velocity?
(1) Speed (2) Velocity
(3) Acceleration (4) Position vectors

PMP358

42. The distance travelled by an object is directly proportional to the time taken. Its acceleration
(1) Increases (2) decreases
(3) becomes zero (4) remains constant

PMP359

43. The distance travelled by an object is directly proportional to the time taken, its speed
(1) Increases (2) decreases
(3) becomes zero (4) remains constant

PMP360

44. If the velocity-time graph of an object is a straight line sloping downwards, the body has
(1) zero acceleration
(2) positive acceleration
(3) constant acceleration
(4) negative acceleration

PMP361

45. If s represent distance and S represent displacement, then $\frac{|S|}{s}$ is
(1) > 1 (2) < 1 (3) = 1 (4) ≤ 1

PMP362

46. When a body is dropped from a tower, then there is an increase in its
(1) mass (2) velocity
(3) acceleration (4) potential energy

PMP363

47. If the displacement-time graph of an object is parallel to the time-axis, then it represents that the object is : -
(1) at rest
(2) in uniform motion
(3) in acceleration motion
(4) none of the above

PMP364

48. The velocity-time graph of motion of an object starting from rest with uniform acceleration is a straight line
(1) parallel to time axis
(2) parallel to velocity axis
(3) passing through origin
(4) none of the above

PMP365

49. If the displacement of a given body is found to be directly proportional to the cube of the time elapsed, then the magnitude of the acceleration of the body is
(1) zero
(2) constant but not zero
(3) increasing with time
(4) decreasing with time

PMP366

50. The acceleration of a moving object can be found from
(1) area under displacement-time graph
(2) slope of displacement-time graph
(3) area under velocity-time graph
(4) slope of velocity-time graph

PMP367

51. Two particles start from rest simultaneously and are equally accelerated. Throughout the motion the relative velocity of one w.r.t. other is : -
(1) zero
(2) non-zero and directed parallel to acceleration
(3) non-zero and directed opposite to acceleration
(4) directed perpendicular to the acceleration

PMP368

52. The relative velocity of a particle moving with a velocity V w.r.t. itself is
(1) V (2) –V
(3) zero (4) None of the above
PMP369

53. A stone is thrown upward and it rises to a height of 100m. The relative velocity of the stone w.r.t. the earth will be maximum at.
(1) the ground (2) a height of 50m
(3) a height of 5m (4) the highest point
PMP370

54. Two particles are moving with velocities V_1 & V_2. Their relative velocity is maximum when the angle between their velocities
(1) zero (2) $\pi/2$ (3) π (4) $\pi/4$
PMP371

55. A body makes a displacement of 4m due East from a point 0 and then makes displacement of 3m due North. Its resultant displacement from O.
(1) 7m (2) 1m (3) 5m (4) 1.2m
PMP372

56. The total flight requires a time that is?
(1) four time the time necessary to reach the maximum height
(2) thrice the time necessary to reach the maximum height
(3) equal to the time necessary to reach the maximum height
(4) twice the time necessary to reach the maximum height
PMP373

57. The angle of projection, for which the horizontal range and the maximum height of a projectile and equal is : -
(1) 45° (2) $\theta = \tan^{-1}(4)$
(3) $\theta = \tan^{-1}(0.25)$ (4) None of these
PMP374

58. For an object thrown at 45° to horizontal, the maximum height (H) and horizontal range (R) are related as
(1) R = 16 H (2) R = 8H
(3) R = 4H (4) R = 2H
PMP375

59. Two bullets are fired horizontally with different velocities from the same height. Which will reach the ground first?
(1) Slower one
(2) Faster one
(3) Both will reach simultaneously
(4) It cannot be predicted
PMP376

60. At the top of the trajectory of a projectile, the acceleration is
(1) maximum (2) minimum
(3) zero (4) g
PMP377

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	3	1	2	3	4	4	1	2	2	2	2	3	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	4	1	3	2	3	3	1	4	4	1	2	3	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	1	1	2	3	3	4	3	2	4	4	3	4	4	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	3	3	4	1	3	1	3	3	4	2	3	3	4

Laws of Motion and Friction

1. Consider the following statements :-

(A) Rolling friction can be smaller than static or sliding friction even by 2 or 3 times of magnitude.

(B) Lubricants are way of reducing static friction in a machine.

(C) Friction involved in ball bearings and surface in contact is rolling friction.

(D) A thin cushion of air maintained between solid surfaces in contact reduces friction.

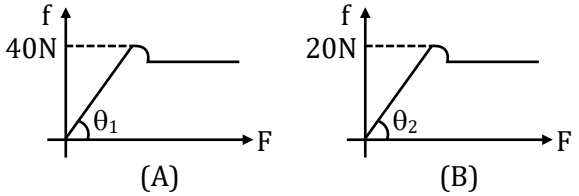
The correct statements from above are :-

(1) A, B, C (2) A, B, C, D

(3) A, C, D (4) B, C, D

PLM319

2. If the two curves drawn between frictional force f and applied force F for 2 bodies on ground. Then correct statement is :-



(1) $\theta_1 > \theta_2$

(2) Coefficient of static friction may be more for (B)

(3) Coefficient of kinetic friction is same as coefficient of static friction for both (A) & (B)

(4) Coefficient of static friction will be more for (A)

PLM320

3. A student unable to answer a question on Newton's laws of motion attempts to pull himself up by tugging on his hair. He will not succeed :-

(1) as the force exerted is small

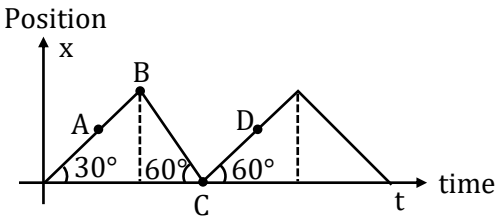
(2) the frictional force while gripping is small

(3) Newton's law of inertia is not applicable to living beings

(4) as the force applied is internal to the system

PLM321

4.



The correct ascending order of magnitude of impulse at points A, B, C, D is :-

(1) A < B < C < D (2) A = B < C < D

(3) A = D < B < C (4) A = D < C < B

PLM322

5. If $v = p + qt^2 + rt^3$. Then net force acting on this particle will be zero at t = 2 if :-

(1) $\left|\frac{p}{q}\right| = 0$ (2) $\frac{q}{r} = 3$

(3) $\left|\frac{r}{q}\right| = \frac{1}{3}$ (4) $\left|\frac{r}{q}\right| = \frac{-1}{3}$

PLM323

6. According to Galileo's experiment for a double incline plane, if slope of second plane is decreased and the planes are smooth, then a ball is released from rest on one of the planes rolls down and climb up the other the final height of the ball is :-

(1) Less than initial height

(2) More than initial height

(3) Equal to initial height

(4) More or less than initial height

PLM324

7. No force acts on particle, it is found that particle has non-zero acceleration. Then :-

(1) Frame of reference is inertial

(2) Frame of reference is non-inertial

(3) Newton's first law is applicable in this situation

(4) None of the above

PLM325

8. Action and reaction are equal and opposite. If F be the magnitude of both the action and reaction, then :-

(1) Resultant of action and reaction is zero on a body.

(2) Resultant of action and reaction is 2F

(3) Resultant of action and reaction is less than F

(4) Action and reaction do not cancel each-other

PLM326

9. A box is placed on the surface of a truck. When the truck accelerates in the forward direction, then the direction of force of friction on lower surface of the box due to surface of the truck to prevent the sliding of the box is :-
(1) In backward direction
(2) In forward direction
(3) In upward direction
(4) In downward direction

PLM327

10. A car accelerates on a horizontal road due to the force exerted by :-
(1) The road
(2) The driver of car
(3) The engine of the car
(4) The tyre

PLM328

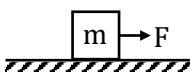
11. Which of the following statements is not true?
(1) The same force is required to act for same time for the same change in momentum for different bodies of same mass
(2) The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts
(3) A greater opposing force is needed to stop a heavy body than a light body in the same time, if they are moving with the same speed
(4) The greater is the change in the momentum in a given time, the lesser is the force that needs to be applied

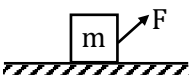
PLM329

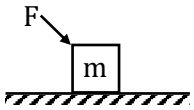
12. μ_s, μ_k and μ_r are the coefficients of static friction, kinetic friction and rolling friction respectively then :-
(1) $\mu_s > \mu_k > \mu_r$ (2) $\mu_s > \mu_k < \mu_r$
(3) $\mu_s < \mu_k > \mu_r$ (4) $\mu_s < \mu_k < \mu_r$

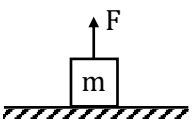
PLM330

13. A block of mass m is placed at rest on a rough horizontal surface with the coefficient of friction μ. The following figure shows four different ways in which a constant force F may be applied on the block. Identify the situation in which the block can be moved with the least effort :-

(1) 

(2) 

(3) 

(4) 

PLM331

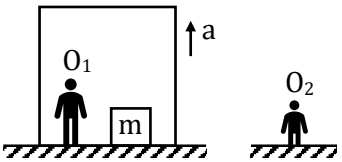
14. A particle is moving with a constant speed along a straight line path. A force is not required to:
(1) increase its speed
(2) decrease its momentum
(3) change the direction
(4) keep it moving with uniform velocity

PLM332

15. An object will continue accelerating until :-
(1) Resultant force on it begins to decrease
(2) Its velocity changes direction
(3) The resultant force on it is zero
(4) The resultant force is at right angles to its direction of motion

PLM333

16. Observer O_1 is in a lift going upward and O_2 is on ground both apply Newton's law and measure normal reaction on the body :-



(1) The both measure the same value
(2) The both measure different value
(3) The both measure zero
(4) Not sufficient data

PLM334

17. A cold soft drink is kept on the balance. When the cap is open, then the weight :-
(1) Increases
(2) Decreases
(3) First increases then decreases
(4) Remains same

PLM335

18. Inertia is the property of a body linked to tendency of a body
(1) to change its position
(2) to change its direction
(3) to change the momentum
(4) to resist any change in the state

PLM336

19. Newton's second and third laws of motion lead to the conservation of
(1) Linear momentum
(2) Angular momentum
(3) Potential energy
(4) Kinetic energy

PLM337

20. A body whose momentum is constant must have constant
(1) velocity (2) force
(3) acceleration (4) All of the above

PLM338

21. A cannon after firing recoils due to
(1) conservation of energy
(2) backward thrust of gases produced
(3) Newton's third law
(4) Newton's first law

PLM339

22. The mass of a body is 2kg. It weight is
(1) 19.6N (2) 20N (3) 30N (4) 40N

PLM340

23. The mass of a body which is equal to the ratio of force acting on a body to the acceleration produced in the body is
(1) gravitational mass
(2) electromagnetic mass
(3) Internal mass
(4) Inertial mass

PLM341

24. A passenger in a moving bus is thrown forward when the bus is suddenly stopped. This is explained: -
(1) by Newton's first law
(2) by Newton's second law
(3) by Newton's third law
(4) by the principle of conservation of momentum

PLM342

25. A body of mass 5 kg is travelling with a uniform velocity of 2 m/s. Its momentum is
(1) 10 kg m/s (2) 7 kg m/s
(3) 2.5 kg m/s (4) 3 kg m/s

PLM343

26. When an external force is not applied to the system, its total momentum:-
(1) Becomes zero
(2) Remains constant
(3) Increases gradually
(4) Decreases gradually

PLM344

27. Force acting for a short duration are called
(1) Short force (2) Interval force
(3) Impulsive force (4) Interrupting force

PLM345

28. An aircraft executes a horizontal loop at a speed of 720 km/h with its wings banked at 15°. What is the radius of the loop?
(1) 15 km (2) 23.567m
(3) 12.781m (4) 14 km

PLM346

29. If a stone is thrown out of an accelerated train, then acceleration of the stone at any instant depends on: -
(1) force acting on it at that instant
(2) acceleration of the train
(3) both (1) & (2)
(4) none of these

PLM347

30. Identify the correct statement from the following: -
(1) friction does not depend on the area in contact
(2) friction is always less than the applied force
(3) friction depends on the volume of body
(4) friction depends on the area of contact

PLM348

31. A cyclist bends while taking turn in order to: -
(1) provide required centripetal force
(2) reduce apparent weight
(3) reduce speed
(4) reduce friction

PLM349

32. There are two statements: -
Statement A: - Rate of change of momentum corresponds to force.
Statement B: - Rate of change of momentum corresponds to kinetic energy.
Which one of the following correct?
(1) A only
(2) B only
(3) Both A & B are correct
(4) Both A & B are wrong

PLM350

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	4	3	3	3	2	4	2	1	4	1	2	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	3	4	1	1	3	1	4	1	1	2	3	1	1	1
Question	31	32													
Answer	1	1													

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Work, Energy and Power

1. When a man pushes a wall he gets tired because of :
(1) Internal energy utilised in contracting and extending muscles
(2) By doing positive work on wall
(3) By doing negative work on wall
(4) None

PWP189

2. If a block of mass M slides on a rough table by a horizontal string :
(1) Work done by tension is equal to change in K.E.
(2) Work done by friction is equal to change in K.E.
(3) Work done by friction converts to dissipative losses
(4) Work done by tension is equal to change in P.E.

PWP190

3. Choose incorrect statement :
(1) Energy stored in spring is same when it is stretched or compressed by same length
(2) Power of centripetal force is always zero in circular motion
(3) During inelastic collision of two masses K.E. lost during collision recovers partially
(4) Work done by gravity in projectile motion is always positive

PWP191

4. Choose correct statement :
(1) Slope of force-displacement graph is work done
(2) Area of potential energy-position graph is force
(3) Slope of work-time graph is power
(4) Area of acceleration-position graph is work-done

PWP192

5. Which of the following statements is incorrect?
(1) If momentum of a particle is zero then kinetic energy must be zero
(2) If kinetic energy of a particle is zero then its momentum must be zero
(3) If momentum of a system is zero, then its kinetic energy must be zero
(4) If kinetic energy of a system is zero, then its momentum must be zero

PWP193

6. A Weightlifter lifts a weight off the ground and holds it up
(1) Work is done in lifting as well as holding the weight
(2) No work is done in both lifting and holding the weight
(3) Work is done in lifting the weight but no work is required to be done in holding it up
(4) No work is done in lifting the weight but work is required to be done in holding it up

PWP194

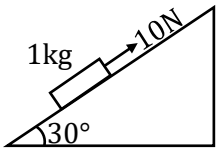
7. A uniform chain of length 2 m is kept on a table such that a length of 60 cm hanges freely from the edge of the table. The total mass of the chain is 4 kg. The work done in pulling the entire chain on the table (Take g = 10 ms^{-2})
(1) 12.9 J (2) 6.3 J
(3) 3.6 J (4) 2.01 J

PWP195

8. A block of mass 2 kg initially at rest moves under the action of an applied horizontal force of 6 N on a rough horizontal surface. The coefficient of friction between block and surface is 0.1. The work done by the applied force in 10 s is (Take g = 10 ms^{-2})
(1) 200 J (2) – 200 J
(3) 600 J (4) – 600 J

PWP196

9. A block of mass 1 kg is pushed up on a surface inclined to horizontal at an angle of 30° by a force of 10 N parallel to the inclined surface as shown in the figure.



The coefficient of friction between block and the inclined is 0.1. If the block is pushed up by 10 m along the incline, then the work against gravity is (Take g = 10 ms^{-2})
(1) 10 J (2) 50J
(3) 100J (4) 150J

PWP197

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10. Which of the following statements is incorrect?
(1) Kinetic energy may be zero, positive or negative
(2) Power, energy and work are all scalars
(3) Potential energy may be zero, positive or negative
(4) Ballistic pendulum is a device for measuring the speed of bullets

PWP198

11. If the force acting on a body is inversely proportional to its speed, then its kinetic energy is
(1) Linearly related to time
(2) Inversely proportional to time
(3) Inversely proportional to the square of time
(4) A constant

PWP199

12. The momentum of a body is increased by 25%. The kinetic energy is increased by about
(1) 25% (2) 5% (3) 56% (4) 38%

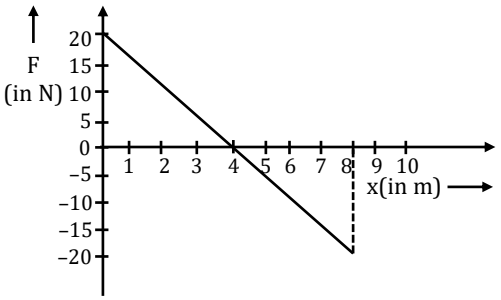
PWP200

13. Two bodies A and B have masses 20 kg and 5 kg respectively. Each one is acted upon by a force of 4 kg wt. If they acquire the same kinetic energy in times t_A and t_B, starting from rest then the ratio $\frac{t_A}{t_B}$ is :-

(1) $\frac{1}{2}$ (2) 2 (3) $\frac{2}{5}$ (4) $\frac{5}{6}$

PWP201

14. A force F acting on an object varies with distance x as shown in the figure.



The work done by the force in moving the object from x = 0 to x = 8 m is
(1) 0 J (2) 80 J (3) –40 J (4) 40 J

PWP202

15. A block of mass 10 kg is moving in x-direction with a constant speed of 10 ms^{-1}. It is subjected to a retarding force $F_x = -0.1x$ N during its travel from x = 20 m to x = 30 m. Its kinetic energy will be :-
(1) 250 J (2) 275 J
(3) 450 J (4) 475 J

PWP203

16. A particle in a certain conservative force field has a potential energy given by $V = \frac{20xy}{z}$. The force exerted on it is :-

(1) $\left(\frac{20y}{z}\right)\hat{i}+\left(\frac{20x}{z}\right)\hat{j}+\left(\frac{20xy}{z^2}\right)\hat{k}$

(2) $-\left(\frac{20y}{z}\right)\hat{i}-\left(\frac{20x}{z}\right)\hat{j}+\left(\frac{20xy}{z^2}\right)\hat{k}$

(3) $-\left(\frac{20y}{z}\right)\hat{i}-\left(\frac{20x}{z}\right)\hat{j}-\left(\frac{20xy}{z^2}\right)\hat{k}$

(4) $\left(\frac{20y}{z}\right)\hat{i}+\left(\frac{20x}{z}\right)\hat{j}-\left(\frac{20xy}{z^2}\right)\hat{k}$

PWP204

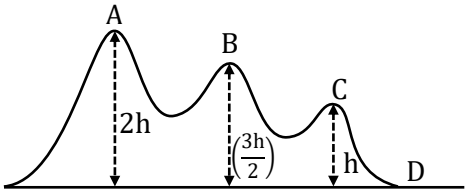
17. A raindrop of mass 1 g falling from a height of 1 km hits the ground with a speed of 50 ms^{-1}. Which of the following statements is correct ? (Take g = 10 ms^{-1})
(1) The loss of potential energy of the drop is 10 J
(2) The gain in kinetic energy of the drop is 1.25 J
(3) The gain in kinetic energy of the drop is not equal to the loss of potential energy of the drop
(4) All of these

PWP205

18. In a shotput event in athlete throws the shotput of mass 20 kg with an initial speed of 2 ms^{-1} at 45° from a height 3 m above ground. Assuming air resistance to be negligible and acceleration due to gravity to be 10 ms^{-2}, the kinetic energy of the shotput when it just reaches the ground will be :-
(1) 2.5 J (2) 5 J
(3) 525 J (4) 640 J

PWP206

19. A small roller coaster starts at point A with a speed u on a curved track as shown in the figure.



The friction between the roller coaster and the track is negligible and it always remains in contact with the track. The speed of roller coaster at point D on the track will be

(1) $(u^2 + gh)^{1/2}$ (2) $(u^2 + 2gh)^{1/2}$

(3) $(u^2 + 4gh)^{1/2}$ (4) u

PWP207

20. The bob of a pendulum is released from a horizontal position. If the length of pendulum is 2 m. What is the speed with which the bob arrives at the lower most point. Assume that 10% of its energy is dissipated against air resistance (Take g = 10 ms^{-2})

(1) 4 m s^{-1} (2) 6 m s^{-1}

(3) 8 m s^{-1} (4) 10 m^{-1}

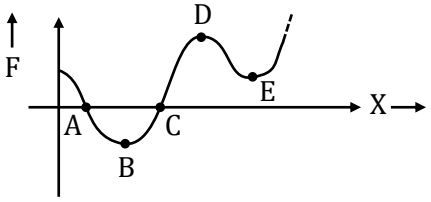
PWP208

21. A ball of mass m is dropped from a cliff of height H. The ratio of its kinetic energy to the potential energy when it is fallen through a height 3/4 H is

(1) 3 : 4 (2) 4 : 3 (3) 1 : 3 (4) 3 : 1

PWP209

22. Force on a body varies with position as shown in figure. It is in unstable equilibrium at point.



(1) D (2) A (3) E (4) C

PWP210

23. Two springs of spring constants 1000 N m^{-1} and 2000 N m^{-1} are stretched with same force. They will have potential energy in the ratio of

(1) 2 : 1 (2) 4 : 1

(3) 1 : 2 (4) 1 : 4

PWP211

24. A 15 g ball is shot from a spring gum whose spring has a force constant 600 N m^{-1}. The spring is compressed by 5 cm. The greatest possible horizontal range of the ball for this compression (Take g = 10 m s^{-2}) is :-

(1) 6 m (2) 8 m (3) 10 m (4) 12 m

PWP212

25. A car of mass 1000 kg moving with a speed 18 km/h on a smooth road collides with a horizontally mounted spring of spring constant 6.25×10^3 N m^{-1}. The maximum compression of the spring is :-

(1) 1 m (2) 2 m (3) 3 m (4) 4 m

PWP213

26. A man weighing 60 kg climbs up a staircase carrying a load of 20 kg on his head. The stair case has 20 steps each of height 0.2 m. If the takes 10 s to climb. Find his power.

(1) 313.6 W (2) 120.6 W

(3) 510 W (4) 0

PWP214

27. A force $(4\hat{i}+\hat{j}-2\hat{k})$ N acting on a body maintains its velocity at $(2\hat{i}+2\hat{j}+3\hat{k})ms^{-1}$. The power exerted is :-

(1) 4 W (2) 5 W (3) 2 W (4) 8 W

PWP215

28. A body is initially at rest. It undergoes one dimensional motion with constant acceleration. The power delivered to it at time t is proportional to :-

(1) $t^{1/2}$ (2) t (3) $t^{3/2}$ (4) t^2

PWP216

29. An elevator which can carry a maximum load of 1800 kg (elevator + passengers) is moving up with a constant speed of 2 ms^{-1}. The frictional force opposing the motion is 4000 N. What is minimum power delivered by the motor to the elevator?

(1) 22 kW (2) 44 kW

(3) 66 kW (4) 88 kW

PWP217

30. Water is flowing in a river at 2 m s^{-1}. The river is 50 m wide and has an average depth of 5 m. The power available from the current in the river is (Density of water = 1000 kg m^{-3})
(1) 0.5 MW (2) 1 MW
(3) 1.5 MW (4) 2 MW
PWP218

31. By how much kinetic energy increase if the momentum is increased by 20%.
(1) 55% (2) 66%
(3) 44% (4) 77%
PWP219

32. No work is done if
(1) displacement is zero
(2) force is zero
(3) force & displacement are mutually perpendicular
(4) All of these
PWP220

33. When the force retards the motion of body, the work done is: -
(1) Zero
(2) Negative
(3) Positive
(4) Positive and Negative depending upon the magnitude of force & displacement
PWP221

34. If a light body and heavy body have same K.E. then which one has greater linear momentum?
(1) Lighter body
(2) Heavier body
(3) Both have same momentum
(4) Can't be predicted
PWP222

35. A man pushes a wall and fails to displace it, he does
(1) Negative work
(2) Positive but not maximum work
(3) No work at all
(4) Maximum positive work
PWP223

36. If two uncharged particles are brought near one another the P.E. of the system will: -
(1) Increase (2) Decrease
(3) Remains same (4) Equal to the K.E.
PWP224

37. The potential energy of a system increases if work is done
(1) Upon the system by a non-conservative force
(2) By the system against a conservative force
(3) By the system against a non-conservative force
(4) Upon the system by a conservative force
PWP225

38. A body of mass 10 kg is travelling with uniform speed of 5 m/s. Its kinetic energy is
(1) 25J (2) 125J
(3) 1250J (4) 1000J
PWP226

39. How much water a pump of 2KW can raise in one minute to a height of 10m, take g = 10 m/s^2
(1) 1200 (2) 100
(3) 2000 (4) 1000
PWP227

40. A ball moves in a frictionless inclined table without slipping. The work done by the table surface on the ball is: -
(1) zero (2) positive
(3) negative (4) None of these
PWP228

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	3	4	3	3	3	3	3	2	1	1	3	2	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	4	3	2	4	4	1	3	2	1	1	2	2	2
Question	31	32	33	34	35	36	37	38	39	40					
Answer	3	4	2	2	3	1	2	2	1	1					

Circular Motion

1. The earth (mass = 6×10^{24} kg) revolves around the sun with an angular velocity of 2×10^{-7} radian/sec in a circular orbit of radius 1.5×10^{8} km. The force exerted by the sun, on the earth is :-

(1) 6×10^{19} N (2) 18×10^{25} N

(3) 36×10^{21} N (4) 27×10^{39} N

PCM139

2. Which of the following statements is false for a particle moving in a circle with a constant angular speed -

(1) The velocity vector is tangential to the circle

(2) The acceleration vector is tangential to the circle

(3) The acceleration vector points towards the centre of the circle

(4) The velocity and acceleration vectors are perpendicular to each other

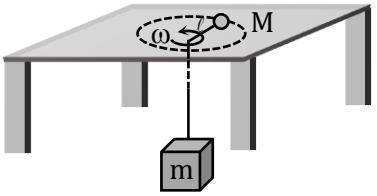
PCM140

3. What is approximately the centripetal acceleration (in units of acceleration due to gravity on earth, g = 10 ms^{-2}) of an air-craft flying at a speed of 400 ms^{-1} through a circular arc of radius 0.6 km?

(1) 26.7 (2) 16.9 (3) 13.5 (4) 30.2

PCM141

4. A block of mass M is situated on a smooth horizontal frictionless table. A thread tied to the block passes through a hole in the table and carries a mass m at its other end. If the length of the thread above the table is ℓ, what should be the value of m so that it may remain suspended at a constant height and the block M moves in a circular path with an angular velocity ω on the table?

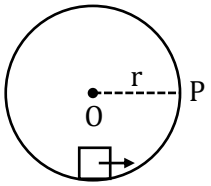


(1) $\frac{M\ell\omega^2}{g}$ (2) $\frac{M\ell\omega^2}{3g}$

(3) $\frac{M\ell\omega^2}{5g}$ (4) $\frac{2M\ell\omega^2}{g}$

PCM142

5. A particle is projected with velocity just to complete the loop. At position P the speed of particle is :–

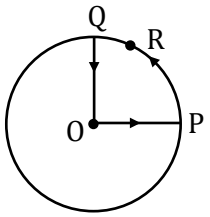


(1) $\sqrt{5gr}$ (2) $\sqrt{3gr}$

(3) $\sqrt{2gr}$ (4) $\sqrt{4gr}$

PCM143

6. A cyclist starts from centre O of a circular park of radius 1 km and moves along the path OPRQO as shown in figure. If he maintains constant speed of 10 ms^{-1}, what is his acceleration at point R?



(1) 10 ms^{-2} (2) 0.1 ms^{-2}

(3) 0.01 ms^{-2} (4) 1 ms^{-2}

PCM144

7. A particle tied with a string moves in a vertical circular path. If X_1 and X_2 are the extreme left and right positions of the particle on the path, whereas Y_1 and Y_2 the extreme upper and lower positions, then the tension(s) in the string at the position(s).

(1) X_1 is greater than that at X_2

(2) Y_2 is greater than that at Y_1

(3) Y_1 and Y_2 are the same

(4) Y_1, Y_2, X_1 and X_2 are equal

PCM145

8. A motor cycle is going on an over bridge of radius 'R'. The dirver maintains a constant speed. As the motor cycle is ascending on the overbridge, the normal force on it.

(1) Increases

(2) Decreases

(3) Remains the same

(4) Fluctuates erratically

PCM146

9. Two particles of equal masses are revolving in circular paths of radii r_1 and r_2 respectively with the same time period. The ratio of their centripetal force is :

(1) r_1/r_2 (2) $\sqrt{r_2/r_1}$

(3) $(r_1/r_2)^2$ (4) $(r_2/r_2)^2$

PCM147

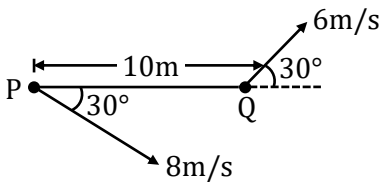
10. A body of mass 1 kg is rotating in a vertical circle of radius 1 m with critical speed. What will be the difference in its kinetic energy at the top and bottom of the circle?

($g = 10\ ms^{-2}$)

(1) 10 J (2) 20 J (3) 30 J (4) 50 J

PCM148

11. Two moving particles 'P' and 'Q' are 10 m apart at an certain instant. The velocity of P is 8 m/s making an angle 30° with line joining P and Q and velocity of Q is 6 m/s making an angle 30° with PQ as shown in figure. Then the angular speed of P with respect to Q is –



(1) 0 rad/s (2) 0.1 rad/s

(3) 0.4 rad/s (4) 0.7 rad/s

PCM149

12. A heavy small sized sphere is suspended by a string of length ℓ. The sphere rotates uniformly in a horizontal circle with the string making an angle θ with the vertical. Then the time period of this conical pendulum is :

(1) $T = 2\pi\sqrt{\frac{\ell}{g}}$ (2) $T = 2\pi\sqrt{\frac{\ell\sin\theta}{g}}$

(3) $T = 2\pi\sqrt{\frac{\ell\cos\theta}{g}}$ (4) $T = 2\pi\sqrt{\frac{\ell}{g\cos\theta}}$

PCM150

13. A body moves in a circular path of radius r = 450 m with tangential acceleration $a_t = 2\ m/s^2$. When its tangential linear velocity is 30 m/s, the total acceleration will be :

(1) $5.4\ ms^{-2}$ (2) $3.9\ ms^{-2}$

(3) $2.8\ ms^{-2}$ (4) $2.1\ ms^{-2}$

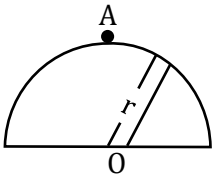
PCM151

14. A body of mass 4 kg is being rotated with 120 rev per minute in a horizontal circular path of radius $\frac{2}{\pi}$m . Its kinetic energy is :

(1) 2 J (2) 32 J (3) 80 J (4) 128 J

PCM152

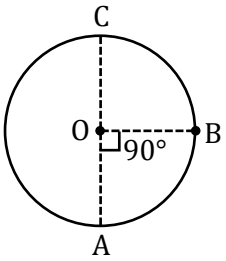
15. A mass m slides from rest down the surface of a frictionless hemispherical bowl of radius r from the highest point A. The velocity of mass when it reaches the bottom is :



(1) $\sqrt{2gr}$ (2) $\sqrt{mgr}$ (3) 2mgr (4) gr

PCM153

16. A particle crosses the topmost point C of a vertical circle with critical speed; then the ratio of velocities at point A, B and C is :



(1) 3 : 2 : 1 (2) 5 : 3 : 1

(3) $5^2 : 3^2 : 1^2$ (4) $\sqrt{5} : \sqrt{3} : \sqrt{1}$

PCM154

17. A particle of mass m is being circulated on a vertical circle of radius r. If the speed of particle at the highest point be υ, then :

(1) $mg = \frac{mv^2}{r}$ (2) $mg > \frac{mv^2}{r}$

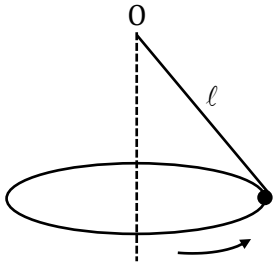
(3) $mg \le \frac{mv^2}{r}$ (4) $mg \ge \frac{mv^2}{r}$

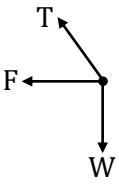
PCM155

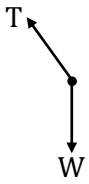
18. A car is moving in a circular horizontal track of radius 10 m with a constant speed of 10 m/sec. A bob is suspended from the roof of the car by a light rigid rod of length 1.00 m. The angle made by the rod with the vertical is:
(1) zero (2) 30° (3) 45° (4) 60°

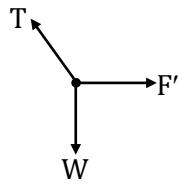
PCM156

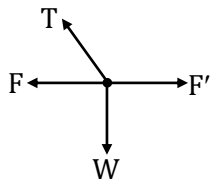
19. A point mass m is suspended from a light thread of length ℓ, fixed at O, is whirled in a horizontal circle at constant speed as shown. For a frame, stationary with respect to the mass, the forces on the mass are :



(1) 

(2) 

(3) 

(4) 

PCM157

20. A particle of mass m is tied to a string of length L and whirled into a horizontal plane. If the tension in the string is T, then the speed of the particle will be: -

(1) $\sqrt{\frac{T\ell}{m}}$ (2) $\sqrt{\frac{2T\ell}{m}}$

(3) $\sqrt{\frac{3T\ell}{m}}$ (4) $\sqrt{\frac{T}{m\ell}}$

PCM158

21. A coin placed on a rotating turntable just slips if it is placed at a distance of 4 cm from the centre. If the angular velocity of the turntable is doubled, it will just slip at a distance of
(1) 1 cm (2) 2 cm
(3) 4 cm (4) 8 cm

PCM159

22. A car moves at a constant speed on a road as shown in figure. The normal force by the road on the car is N_A and N_A when it is at points A & B respectively.



(1) $N_A = N_B$
(2) $N_A > N_B$
(3) $N_A < N_B$
(4) insufficient information

PCM160

23. When a particle moves in a circle with a uniform speed
(1) it's velocity and acceleration are both constant.
(2) it's velocity is constant but acceleration changes.
(3) it's acceleration is constant but velocity changes.
(4) it's velocity and acceleration both change

PCM161

24. A circular road of radius 50m has the angle of banking equal to 30°. At what speed should a vehicle go on this road so that friction is not used
(1) 22 m/s (2) 17 m/s
(3) 10 m/s (4) None of these

PCM162

25. An object follows a curved path. The following quantities may remain constant during the motion.
(a) speed
(b) velocity
(c) acceleration
(d) magnitude of acceleration
(1) (a), (d)
(2) (a), (b) and (c)
(3) (a), (b), (c) and (d)
(4) (a) only

PCM163

AIM: FREE EDUCATION TO ALL APUL

26. A person applies a constant force $\vec{F}$ on a particle of mass m and finds that the particle moves in a circle of radius r with a uniform speed v as seen from an inertial frame of reference.

(a) This is not possible.

(b) There are other forces on the particle.

(c) The resultant of the other forces is $\frac{mv^2}{r}$ toward the center.

(d) The resultant of the other forces varies is magnitude as well as in direction.

(1) a,b,c (2) a,c (3) b,d (4) only d

PCM164

27. The position vector of a particle in a circular motion about the origin sweeps out equal area is equal time. It's

(a) velocity remains constant

(b) speed remains constant

(c) acceleration remains constant

(d) tangential acceleration remains constant

(1) a, c

(2) b only

(3) b, d

(4) a, c and d

PCM165

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	1	1	2	2	2	1	1	2	4	3	3	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27			
Answer	4	3	3	3	1	1	3	4	2	1	3	2			

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Collisions and Centre of Mass

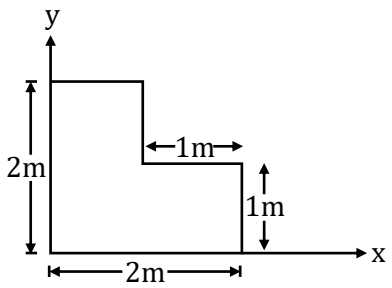
1. When an explosive shell travelling in a parabolic path under the effect of gravity explodes in the mid-air, the centre of mass of the fragments will move: -
(1) vertically downwards
(2) along the original parabolic path
(3) vertically upwards and then vertically downwards
(4) horizontally followed by parabolic path

PSC157

2. The centre of mass of a body: -
(1) lies always at the geometrical centre
(2) lies always inside the body
(3) lies always outside the body
(4) may lie within or outside the body

PSC158

3. The x,y coordinates of the centre of mass of a uniform L-shaped lamina of mass 3 kg is :-



(1) $\left(\frac{5}{6}m,\frac{5}{6}m\right)$ (2) (1m, 1m)

(3) $\left(\frac{6}{5}m,\frac{6}{5}m\right)$ (4) (2m, 2m)

PSC159

4. The center of mass of a system of two particles of masses m_1 and m_2 is at a distance d_1 from m_1 and at a distance d_2 from mass m_2 such that: -

(1) $\frac{d_1}{d_2}=\frac{m_2}{m_1}$ (2) $\frac{d_1}{d_2}=\frac{m_1}{m_2}$

(3) $\frac{d_1}{d_2}=\frac{m_1}{m_1+m_2}$ (4) $\frac{d_1}{d_2}=\frac{m_2}{m_1+m_2}$

PSC160

5. Two particles of mass 1 kg and 3 kg have position vectors $2\hat{i}+3\hat{j}+4\hat{k}$ and $-2\hat{i}+3\hat{j}-4\hat{k}$ respectively. The center of mass has a position vector :

(1) $\hat{i}+3\hat{j}-2\hat{k}$ (2) $-\hat{i}-3\hat{j}-2\hat{k}$

(3) $-\hat{i}+3\hat{j}+2\hat{k}$ (4) $-\hat{i}+3\hat{j}-2\hat{k}$

PSC161

6. The velocity of centre of mass of the system remains constant, if the total external force acting on the system is :
(1) minimum (2) maximum
(3) unity (4) zero

PSC162

7. Two particles of equal mass have velocities $\vec{v}_1=2\hat{i}$ ms^{-1} and $\vec{v}_2=2\hat{j}$ ms^{-1}. First particle has an acceleration $\vec{a}_1=\left(3\hat{i}+3\hat{j}\right)$ ms^{-2} while the acceleration of the other particles is zero then COM of both particles moves in a path of:-
(1) straight line (2) parabola
(3) circle (4) ellipse

PSC163

8. A child is standing at on a long, trolley moving with speed v on a smooth horizontal floor. If the child starts running towards the other end of the trolley with a speed u, the centre of mass of the system (trolley + child) will move with a speed
(1) zero (2) (v + u)
(3) (v – u) (4) v

PSC164

9. A spherical ball of mass m_1 collides head on with another ball of mass m_2 at rest. The collision is elastic. The fraction of kinetic energy lost by m_1 is: -

(1) $\frac{4m_1m_2}{(m_1+m_2)^2}$ (2) $\frac{m_1}{m_1+m_2}$

(3) $\frac{m_2}{m_1+m_2}$ (4) $\frac{m_1m_2}{(m_1+m_2)^2}$

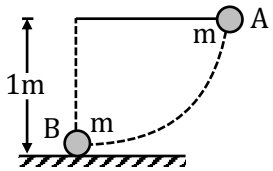
PSC165

10. A ball of mass m moving with a speed $2v_0$ collides head on with an identical ball at rest. If e is the coefficient of restitution, then what will be the ratio of velocity of first ball to second ball after collision?

(1) $\frac{1-e}{1+e}$ (2) $\frac{1+e}{1-e}$ (3) $\frac{e-1}{e+1}$ (4) $\frac{e+1}{e-1}$

PSC166

11. The bob A of a pendulum of mass m released from horizontal to the vertical hits another bob B of the same mass at rest on a table as shown in figure. If the length of the pendulum is 1m, what is the speed with which bob B starts moving. (Neglect the size of the bobs and assume the collision to be elastic)
(Take g = 10 ms^{-2}) :



(1) 4.47 ms^{-1} (2) 5.47 ms^{-1}
(3) 6.47 ms^{-1} (4) 3.47 ms^{-1}

PSC167

12. A spherical ball A of mass 4 kg, moving along a straight line strike another spherical ball B of mass 1 kg at rest. After the collision, A and B move with velocities v_1 ms^{-1} a v_2 ms^{-1} respectively making angles of 30° and 60° with respect to the original direction of motion of A. The ratio $\frac{v_1}{v_2}$ will be :-

(1) $\frac{\sqrt{3}}{4}$ (2) $\frac{4}{\sqrt{3}}$ (3) $\frac{1}{\sqrt{3}}$ (4) $\sqrt{3}$

PSC168

13. The density of a non-uniform rod of length 1m is given by $\rho(x) = a(1 + bx^2)$ where a and b are constant and $0 \le x \le 1$. The centre of mass of the rod will be at :

(1) $\frac{3(2+b)}{4(3+b)}$ (2) $\frac{4(2+b)}{3(3+b)}$
(3) $\frac{3(3+b)}{4(2+b)}$ (4) $\frac{4(3+b)}{3(2+b)}$

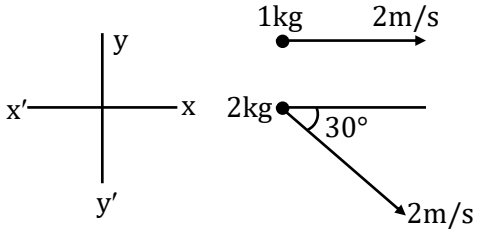
PSC169

14. Masses 8, 2, 4, 2 kg are placed at the corners A, B, C, D respectively of a square ABCD of diagonal 80 cm. The distance of centre of mass from A will be

(1) 20 cm (2) 30 cm
(3) 40 cm (4) 60 cm

PSC170

15. The velocity of centre of mass of the system as shown in the figure



(1) $\left(\frac{2-2\sqrt{3}}{3}\right)\hat{i}-\frac{1}{3}\hat{j}$ (2) $\left(\frac{2+2\sqrt{3}}{3}\right)\hat{i}-\frac{2}{3}\hat{j}$
(3) $4\hat{i}$ (4) None of these

PSC171

16. A ball is thrown vertically downwards from a height of 20 m with an initial velocity v_0. It collides with the ground, loses 50% of its energy in collision and rebounds to the same height. The initial velocity v_0 is:
(Take g = 10 m/s^2)

(1) 10 m/s (2) 14 m/s
(3) 20 m/s (4) 28 m/s

PSC172

17. Two particles whose masses are 10 kg and 30 kg and their position vectors are $\hat{i}+\hat{j}+\hat{k}$ and $-\hat{i}-\hat{j}-\hat{k}$ respectively would have the centre of mass at position: -

(1) $-\frac{\left(\hat{i}+\hat{j}+\hat{k}\right)}{2}$ (2) $\frac{\left(\hat{i}+\hat{j}+\hat{k}\right)}{2}$
(3) $-\frac{\left(\hat{i}+\hat{j}+\hat{k}\right)}{4}$ (4) $\frac{\left(\hat{i}+\hat{j}+\hat{k}\right)}{4}$

PSC173

18. Two bodies of mass 10 kg and 2 kg are moving with velocity $\left(2\hat{i}-7\hat{j}+3\hat{k}\right)$ m/s and $\left(-10\hat{i}+35\hat{j}-3\hat{k}\right)$ m/s respectively. Find the velocity of their centre of mass.

(1) $\hat{k}$ m/s (2) $\left(2\hat{i}+2\hat{k}\right)$ m/s
(3) $\hat{j}$ m/s (4) $2\hat{k}$ m/s

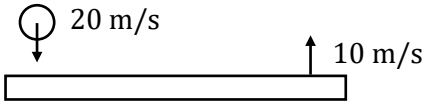
PSC174

19. An explosion blows a rock into three parts. Two parts go off at right angles to each other. These two are, 1 kg first part moving with a velocity of 12 m/s and 2 kg second part moving with a velocity of 8 m/s. If the third part flies off with a velocity of 4 m/s, its mass would be :-

(1) 3 kg (2) 5 kg
(3) 7 kg (4) 17 kg

PSC175

20. A ball hits a surface with a speed of 20 m/s as shown. If surface continues to move with same velocity then find velocity with which ball will rebound if collision is elastic: -



(1) 20 m/s (2) 40 m/s
(3) 30 m/s (4) 50 m/s

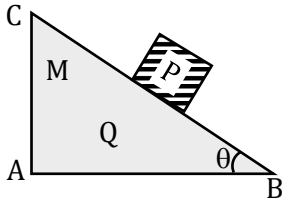
PSC176

21. A bomb is projected with 200m/s at an angle 60° with horizontal. At the highest point, it explodes into three particles of equal masses. One goes vertically upward with velocity 100m/sec, second particle goes vertically downward with the same velocity as the first. Then what is the velocity of the third one-

(1) 120 m/sec with 60° angle
(2) 200 m/sec with 30° angle
(3) 50 m/sec, in horizontal direction
(4) 300 m/sec, in horizontal direction

PSC177

22. A block Q of mass M is placed on a horizontal frictionless surface AB and a body P of mass m is released on its frictionless slope. As P slides by a length L on this slope of inclination θ, the block Q would slide by a distance :-



(1) $\frac{m}{M}L\cos\theta$ (2) $\frac{m}{M+m}L$
(3) $\frac{M+m}{mL\cos\theta}$ (4) $\frac{mL\cos\theta}{m+M}$

PSC178

23. A metal ball of mass 2kg moving with a velocity of 36 km/h has an head on collision with a stationary ball of mass 3kg. If after the collision, the two balls move together, the loss in kinetic energy due to collision is :-

(1) 40 J (2) 60 J
(3) 100 J (4) 140 J

PSC179

24. A neutron collides, head-on with a deuterium at rest. What fraction of the neutron's energy would be transferred to the deuterium?

(1) 89% (2) 11% (3) 79% (4) 21%

PSC180

25. If particles of mass 1kg, 2kg and 3kg are placed at (1, 0, 1), (2, 2, 3) and (1, 3, 0). Position of the centre of mass of the system is :-

(1) $\left(\frac{3}{2},\frac{1}{5},\frac{1}{6}\right)$ (2) $\left(\frac{5}{6},\frac{2}{3},\frac{8}{4}\right)$
(3) $\left(\frac{4}{6},\frac{2}{6},\frac{8}{6}\right)$ (4) None

PSC181

26. Two spheres of masses 2 M and M are initially at rest at a distance R apart. Due to mutual force of attraction they approach each other when they are at separation $\frac{R}{2}$, the acceleration of the centre of mass of sphere would be :

(1) zero (2) g m/s²
(3) 3g m/s² (4) 12g m/s²

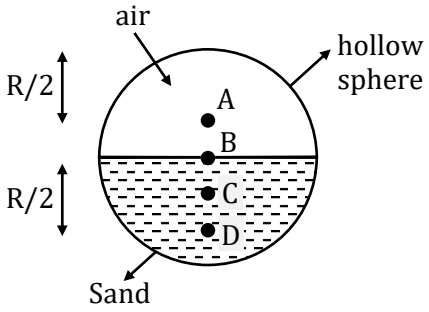
PSC182

27. A ball starts falling from a height h_0. There are n collisions with the earth. If after n collisions the ball rises to a height h_n, then coefficient of restitution e is given by:

(1) $e^n=\sqrt{\frac{h_n}{h_0}}$ (2) $e^n=\sqrt{\frac{h_0}{h_n}}$
(3) $ne=\sqrt{\frac{h_0}{h_n}}$ (4) $\sqrt{ne}=\frac{h_n}{h_0}$

PSC183

28. Which of the following points is likely the position of centre of mass?



(1) A (2) B (3) C (4) D

PSC184

29. Statement-1: The motion of rigid body which is not pivoted or fixed in some way is either a pure translation or a combination of translation and rotation.

Statement-2: The motion of a rigid body which is pivoted or fixed in some way is rotation.

(1) Statement-1 is true; Statement-2 is false

(2) Statement-1 is false; Statement-2 is true

(3) Both statements are true

(4) Both statements are false

PSC185

30. Rigid body is a body with

(1) Perfectly definite shape

(2) Unchanging shape

(3) Distance between all pair of particles of such body do not change

(4) All of the above

PSC186

31. The position of centre of mass of a system of particles does not depend upon the

(1) relative distance between the particles

(2) position of the particles

(3) symmetry of body

(4) mass of particles

PSC187

32. A stick is thrown in the air and lands on the ground at some distance from the thrower. The centre of mass of the stick will move along a parabolic path

(1) in all cases

(2) only if the stick is uniform

(3) only if the stick has linear motion but no rotational motion

(4) only if the stick has a shape such that if it's centre of mass is located at some point on it and not outside it.

PSC188

33. A solid sphere is rotating about a diameter at an angular velocity ω. If it cools so that it's radius reduces to $\frac{1}{n}$ of it's original value, it's angular velocity becomes.

(1) $\frac{\omega}{n}$ (2) $\frac{\omega}{n^2}$

(3) $n\omega$ (4) $n^2\omega$

PSC189

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	1	1	4	4	1	4	1	1	1	1	1	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	4	2	2	4	4	2	1	4	1	1	3	3	4
Question	31	32	33												
Answer	1	1	4												

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Rotational Motion

1. A vehicle is moving with a velocity v on a curved road of width b and radius of curvature R. For counteracting the centrifugal force on the vehicle, the difference in elevation required in between the outer and inner edges of the road is: -

(1) $\frac{v^2b}{Rg}$ (2) $\frac{vb}{Rg}$

(3) $\frac{vb^2}{Rg}$ (4) $\frac{vb}{R^2g}$

PRM261

2. A particle tied with a string moves in a vertical circular path. If X_1 and X_2 are the extreme left and right positions of the particle on the path, whereas Y_1 and Y_2 the extreme upper and lower positions, then the tension(s) in the string at the position(s).

(1) X_1 is greater than that at X_2

(2) Y_2 is greater than that at Y_1

(3) Y_1 and Y_2 are the same

(4) Y_1, Y_2, X_1 and X_2 are equal

PRM262

3. The ratio of the radii of gyration of a circular disc and a circular ring of the same radius about a tangential axis perpendicular to plane of disc or ring is: -

(1) 1 : 2 (2) $\sqrt{5}:\sqrt{6}$

(3) 2 : 3 (4) $\frac{\sqrt{3}}{2}$

PRM263

4. Which of the following statements are correct about a rigid body -

(a) Ideally a rigid body has a perfectly definite and unchanging shape.

(b) Distances between different pairs of particle of rigid body do not change

(c) Inter molecular forces for a rigid body are weak

(d) No real body are truly rigid

(1) a, b, c

(2) a, b, d

(3) c,b,d

(4) All of the above are correct

PRM264

5. Select the incorrect option: -

(1) The total momentum of a system of particles is equal to the product of the total mass of the system and the velocity of its centre of mass.

(2) In pure rotation all parts of the body have the same angular velocity at any instant of time.

(3) For rotation about a fixed axis, the direction of angular velocity does not change with time if $\tau_{axis} = 0$

(4) For rotation about a fixed axis magnitude of angular velocity also remains constant

PRM265

6. Which of the following is wrong: -

(1) The vanishing of the total external force and the vanishing of the total external torque are independent conditions

(2) We can have net external force without having torque

(3) We can have net torque without having net external force

(4) If a force couple acting on a body net external force is zero and also net torque is zero

PRM266

7. Which of the following is incorrect: -

(1) The centre of gravity of an external body is that point where the total gravitational torque on the body is zero.

(2) In rotational equilibrium the total external torque on the body is zero.

(3) Angular velocity $\vec{\omega}$ is a vector directed along axis of rotation.

(4) In pure translation, velocity of different parts is different.

PRM267

8. In pure rolling motion on ground which of the following is wrong :-

(1) Work done by friction is zero

(2) Friction may be directed along the direction of motion

(3) Relative motion of the lowest point w.r.t. ground is zero

(4) Velocity of each point is same as that of centre of mass

PRM268

9. Which of the following is not an example of conservation of angular momentum :-
(1) When a person sitting on turn table suddenly folds his arms, angular velocity increases
(2) Bullet fired from gun
(3) If ice on poles of earth melts then duration of the day increases
(4) Ballet dancer stretches his arms so angular velocity of rotation decreases

PRM269

10. Select the incorrect statement -
(1) The total torque on a system is independent of the origin if the total external force is zero
(2) The centre of gravity of a body coincides with its centre of mass only if the gravitational field does not vary from one part to another of the body.
(3) The angular momentum $\vec{L}$ and the angular velocity $\vec{\omega}$ should be necessarily parallel vectors.
(4) None of these

PRM270

11. If a body rolls down an inclined plane without slipping -
(1) It possess only translatory kinetic energy
(2) It possess only Rotational kinetic energy
(3) No work is done by friction
(4) Gravity does no work

PRM271

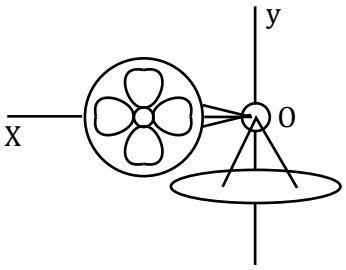
12. Which of the following is wrong -
(1) Radius of Gyration of a body, about an axis may be defined as the distance from the axis of a point mass whose mass is equal to the mass of the given body and moment of inertia is equal to the moment of inertia of body about same axis
(2) Moment of inertia depends on mass of a rigid body, its shape and size
(3) Moment of inertia depends on the distribution of mass about the axis
(4) Moment of inertia is independent of density of body

PRM272

13. Ideally a rigid body is a body with a perfectly definite and unchanging shape :-
(1) Distance between its two points can be changed
(2) No real body is truly rigid
(3) Rigid body has infinite inertia
(4) Rigid body can not perform translational motion

PRM273

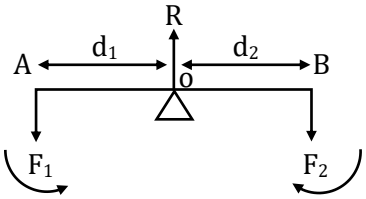
14. Which is axis of precession in the given figure :-



(1) OX (2) OY
(3) both OX and OY (4) None

PRM274

15.

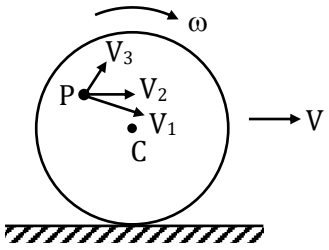


The lever force F_1 is usually some weight to be lifted. It is called the load and its distances from the fulcrum d_1 is called the load arm Force F_2 is the effort applied to lift the load; distance d_2 of the effort from fulcrum then mechanical advantage will be :-

(1) $\frac{F_2}{F_1}$ (2) $\frac{F_1}{F_2}$ (3) F_1F_2 (4) $\frac{1}{F_1F_2}$

PRM275

16. Direction of resultant velocity of particle P in rolling motion :-



Pure Rolling

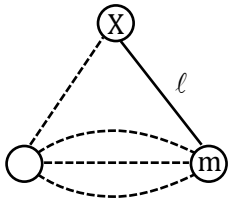
(1) Along V_1 (2) Along V_2
(3) Along V_3 (4) None

PRM276

17. A particle is moving along a straight line parallel to x-axis with constant velocity. Its angular momentum about origin.
(1) Increases with time
(2) Decreases with time
(3) Is constant
(4) Only direction is constant but magnitude will change

PRM277

18. A particle is tied with string and moving in a horizontal plane with constant speed. Its angular momentum about hinge.



(1) Conserved
(2) Only direction is conserved not the magnitude
(3) Only magnitude is conserved not the direction
(4) Neither magnitude nor direction is conserved

PRM278

19. A rod of mass m is released on smooth horizontal surface making angle with horizontal. Then which of the following is incorrect :-
(1) Acceleration of rod along vertical is less than g.
(2) Angular acceleration of rod is not constant
(3) The angular momentum of the rod will not be constant
(4) The torque acting on the rod will be constant

PRM279

20. Which of the following body has equal amount of rotational and translational kinetic energy during pure rolling?
(1) Hollow Cylinder
(2) Solid Cylinder
(3) Hollow Sphere
(4) Solid Sphere

PRM280

21. In a spinning top axis moves around the vertical through its point of contact with the ground sweeping out a cone. This movement of axis of the top around the vertical is known as: -
(1) Rotation (2) Translation
(3) Precession (4) Rolling

PRM281

22. The direction of the angular velocity vector is along: -
(1) the tangent to the circular path
(2) the inward radius
(3) the outward radius
(4) the axis of rotation

PRM282

23. A couple produces: -
(1) Purely translation motion
(2) Purely rotational motion
(3) Both translational and rotational motion
(4) No motion

PRM283

24. Angular momentum of the particle rotating with a central force is constant due to
(1) constant torque
(2) constant force
(3) constant linear momentum
(4) zero torque

PRM284

25. A rigid body is said to be in partial equation when it is in .....
(1) Translational equilibrium only
(2) Rotational equilibrium only
(3) Either (1) or (2)
(4) Neither (1) nor (2)

PRM285

26. Which of the following relations is correct: -
(1) Mechanical advantage = $\frac{\text{Effort}}{\text{Load}}$
(2) Load arm × Effort = Effort arm × Load
(3) Load arm × Load = Effort arm × Effort
(4) None of these

PRM286

27. The MOI of a body depends upon: -
(1) Mass of body
(2) Axis of rotation of body
(3) Shape and size of the body
(4) All of these

PRM287

28. Match the column I with column II

Column-I		Column-II	
(A)	For translational equilibrium	(P)	MK^2
(B)	For Rotational Equilibrium	(Q)	Angular acc.
(C)	MOI of body	(R)	$\Sigma\vec{F}=0$
(D)	Torque is required to produce	(S)	$\Sigma\vec{\tau}=0$

(1) A → p ; B → q ; C → r ; D → s
(2) A → q ; B → r ; C → s ; D → p
(3) A → r ; B → q ; C → p ; D → s
(4) A → r ; B → s ; C → p ; D → q

PRM288

29. A flywheel rotating at 420 rpm slows down at constant rate of 2 rad/s^2. The time required to stop the flywheel is: -
(1) 22s (2) 11s
(3) 44s (4) 12s

PRM289

30. An automobile engine develops 100 KW when rotating at a speed of 1800 rpm. The torque delivered by the engine is: -
(1) $\frac{10^2}{6\pi}$N.m (2) $\frac{10^4}{6\pi}$N.m
(3) $\frac{10^6}{6\pi}$N.m (4) $\frac{10^8}{6\pi}$N.m

PRM290

31. Which of the following principles a circus acrobat employs in his performance.
(1) conservation of energy
(2) conservation of linear momentum
(3) conservation of mass
(4) conservation of angular momentum

PRM291

32. A child is standing with folded hands at the centre of platform rotating about its central axis. The K.E. of the system is K. Now, the child stretches his arms so that MOI of the system doubled. Now the K.E. of the system is : -
(1) K/4 (2) K/2
(3) 2K (4) 4K

PRM292

33. A solid cylinder rolls up an inclined plane of inclination θ with an initial velocity V. How far does the cylinder go up the plane: -
(1) $\frac{3V^2}{2g\sin\theta}$ (2) $\frac{V^2}{4g\sin\theta}$
(3) $\frac{3V^2}{g\sin\theta}$ (4) $\frac{3V^2}{4g\sin\theta}$

PRM293

34. The solid cylinder is rolling without slipping on a plane having inclination θ and the coefficient of static friction μ_s. The relation between θ and μ_s is: -
(1) $\tan\theta > 3\mu_s$ (2) $\tan\theta \le 3\mu_s$
(3) $\tan\theta < 3\mu_s^2$ (4) None of these

PRM294

35. A body is rolling down an inclined plane. If K.E. of rotation is 40% of K.E. in translatory state then the body is
(1) Ring (2) Cylinder
(3) Hollow ball (4) Solid ball

PRM295

36. A solid cylinder of mass M and radius R rolls without slipping down an inclined plane making an angle θ with horizontal then its acceleration is : -
(1) $\frac{g\sin\theta}{3}$ (2) $\frac{2g\sin\theta}{3}$
(3) $\frac{2g\sin\theta}{5}$ (4) $\frac{2g\sin\theta}{7}$

PRM296

37. In above question the force of friction acting on the cylinder is : -
(1) $\frac{2}{3}mg\sin\theta$ (2) $\frac{mg\sin\theta}{3}$
(3) $\frac{2mg\sin\theta}{5}$ (4) $\frac{2mg\sin\theta}{7}$

PRM297

38. When a disc rotates with uniform angular velocity. Which of the following is not true.

(1) The sense of rotation remains same

(2) The orientation of the axis of rotation remains same

(3) The speed of rotation is non-zero and remains same

(4) The angular acceleration is non-zero and remains same

PRM298

39. **Assertion (A): -** The centre of gravity of a body coincides with centre of mass only if the gravitational field does not vary from one part of the body to the other.

Reason (R): - Centre of gravity is independent of the gravitational field.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true and (R) in NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

PRM299

40. **Assertion (A): -** The MOI of a rigid body reduces to its minimum value. When the axis of rotation passes through its centre of gravity.

Reason (R): - The weight of rigid body always acts through its centre of gravity.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true and (R) in NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

PRM300

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	4	2	4	4	4	4	2	3	3	4	2	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	3	4	1	3	4	2	4	3	3	4	4	1	2
Question	31	32	33	34	35	36	37	38	39	40					
Answer	4	2	4	2	4	2	2	4	3	2					

Gravitation

1. The areal velocity and the angular momentum of a planet are related by –
(Where m is the mass of the planet)

(1) $\frac{\Delta\vec{A}}{\Delta t}=\frac{\vec{L}}{2m}$ (2) $\frac{\Delta\vec{A}}{\Delta t}=\frac{\vec{L}}{m}$

(3) $\frac{\Delta\vec{A}}{\Delta t}=\frac{2\vec{L}}{m}$ (4) $\frac{\Delta\vec{A}}{\Delta t}=\frac{\vec{L}}{\sqrt{2}m}$

PGR217

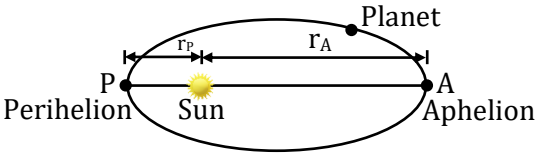
2. A Saturn year is 29.5 times the Earth year. How far is the Saturn from the Sun if the Earth is 1.5×10^8 km away from the Sun?

(1) 1.4×10^6 km (2) 1.4×10^7 km

(3) 1.4×10^8 km (4) 1.4×10^9 km

PGR218

3. A planet orbits the sun in an elliptical path as shown in the figure. Let v_P and v_A be speed of the planet when it is at perihelion and aphelion respectively. Which of the following relations is correct?



(1) $\frac{r_P}{r_A}=\frac{v_A}{v_P}$ (2) $\frac{r_P}{r_A}=\frac{v_P}{v_A}$

(3) $\frac{r_P}{r_A}=\sqrt{\frac{v_P}{v_A}}$ (4) $\frac{r_P}{r_A}=\sqrt{\frac{v_A}{v_P}}$

PGR219

4. Two spheres of masses m and M are situated in air and the gravitational force between them is F. The space around the masses is now filled with a liquid of specific gravity 3. The gravitational force will now be :-

(1) 3F (2) F

(3) $\frac{F}{3}$ (4) $\frac{F}{9}$

PGR220

5. Two identical spheres each of mass M and radius R are separated by a distance 10 R. The gravitational force on mass m placed at the midpoint of the line joining centers of the spheres is :-

(1) Zero (2) $\frac{2GMm}{25R^2}$

(3) $\frac{GMm}{25R^2}$ (4) $\frac{GMm}{100R^2}$

PGR221

6. The ratio of radii of earth to another planet is 2/3 and the ratio of their mean densities is 4/5. If an astronaut can jump to a maximum height of 1.5 m on the earth, with the same effort, the maximum height he can jump on the planet is: -

(1) 1 m (2) 0.8 m

(3) 0.5 m (4) 1.25 m

PGR222

7. The earth (mass = 6×10^{24} kg) revolves around the sun with an angular velocity of 2×10^{-7} radian/sec in a circular orbit of radius 1.5×10^8km. The force exerted by the sun, on the earth is :-

(1) 6×10^{19} N (2) 18×10^{25} N

(3) 36×10^{21} N (4) 27×10^{39} N

PGR223

8. Kepler's law of areas is a consequence of :

(1) Conservation of linear momentum

(2) Net force on planet

(3) Conservation of angular momentum

(4) Constant angular speed

PGR224

9. The total mechanical energy of Earth-Satellite system is :

(1) Positive

(2) Zero

(3) Negative

(4) Can have any value

PGR225

10. If the orbital radius of satellite is decreased, then (PE = potential energy, KE = kinetic energy)
(1) PE decreases and KE increases
(2) Both PE and KE decreases
(3) Both PE and KE increases
(4) PE increases and KE decreases

PGR226

11. A particle of mass m is placed inside a spherical shell of mass M at a point other than the centre :
(1) The particle will move towards the surface of shell
(2) The particle will move towards the centre
(3) The particle will remain stationary
(4) None of the above

PGR227

12. When you go from equator to the poles the value of acceleration due to gravity :
(1) Increases
(2) Decreases
(3) Remains same
(4) Decreases upto a latitude of 45° and then increases

PGR228

13. The radius of the earth is 6400 km and $g = 10\ m/s^2$. In order that a body of 5 kg weighs zero at the equator, the angular speed of the earth is: -
(1) $\frac{1}{80}$ rad / s (2) $\frac{1}{400}$ rad / s
(3) $\frac{1}{800}$ rad / s (4) $\frac{1}{1600}$ rad / s

PGR229

14. A body weighs 500N on the surface of the earth. How much would it weight half way below the surface of the earth?
(1) 1000N (2) 500N
(3) 250N (4) 125N

PGR230

15. The time period of a satellite of earth is 5 hours. If the separation between the earth and the satellite is increased to 4 times the previous value, the new time period will become
(1) 10 hrs (2) 20 hrs
(3) 40 hrs (4) 80 hrs

PGR231

16. What would be the duration of the year if the distance between the earth and the sun gets doubled?
(1) 1032 days (2) 129 days
(3) 365 days (4) 730 days

PGR232

17. The escape velocity for a body projected vertically upwards from the surface of the earth is 11 km/s. If the body is projected at an angle of 45° with the vertical the escape velocity will be
(1) $\frac{11}{\sqrt{2}}$ km / s (2) $11\sqrt{2}$ km/s
(3) 2 km/s (4) 11 km/s

PGR233

18. There is no atmosphere on the moon because
(1) it is closer to the earth
(2) it revolves round the earth
(3) it gets light from the sun
(4) the escape velocity of gas molecules is less than their root mean square velocity here

PGR234

19. The value of g at a particular point is $9.8\ m/s^2$ suppose the earth suddenly shrink uniformly to half its present size without losing any mass. (assuming that the distance of the point from the centre of the earth does not shrink) will become.
(1) $9.8\ m/s^2$ (2) $4.9\ m/s^2$
(3) $19.6\ m/s^2$ (4) $2.45\ m/s^2$

PGR235

20. If the radius of the earth were to be raise by 1% its mass remaining the same, the acceleration due to gravity on the surface of the earth will
(1) increase by 1%
(2) decrease by 2%
(3) decrease by 1%
(4) increase by 2%

PGR236

21. The masses of two planets are in the ratio 1:2. Their radii are in the ratio 1:2. The acceleration due to gravity on the planets are in the ratio.
(1) 1:2 (2) 2:1
(3) 3:5 (4) 5:3

PGR237

22. Which law describes the orbits of planets around the sun?
(1) Newton's law
(2) Faraday's law
(3) Kepler's law
(4) Kirchoff's law

PGR238

23. Newton's law of universal gravitational force is given by
(1) $F = \frac{Gm_1m_2}{r^4}$ (2) $F = \frac{Gm_1m_2}{r^2}$
(3) $F = \frac{Gm_1 + m_2}{r^2}$ (4) $F = \frac{Gm_1 - m_2}{r^2}$

PGR239

24. Force of gravitational attraction is least
(1) at the equator
(2) at the poles
(3) at a point in between equator and any pole
(4) None of these

PGR240

25. In some region, the gravitational field is zero, the gravitational potential in this region.
(1) must be zero
(2) must be constant
(3) must be variable
(4) cannot be zero

PGR241

26. Two satellites of mass m, and 10m are put in same orbit around the sun. If T_1 and T_2 be their time periods, then $\frac{T_1}{T_2} =$
(1) 4 (2) 1
(3) 12 (4) 14

PGR242

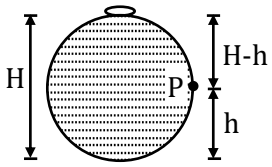
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	1	2	1	2	3	3	3	1	3	1	3	3	3
Question	16	17	18	19	20	21	22	23	24	25	26				
Answer	1	4	4	1	2	2	3	2	1	2	2				

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Properties of Matter and Fluid Mechanics

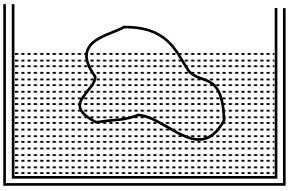
1. A vessel contains a liquid of density ρ as shown in figure. The gauge pressure at the point P is:



(1) hρg (2) Hρg
(3) (H – h)ρg (4) (H – h)ρg cos θ

PPF279

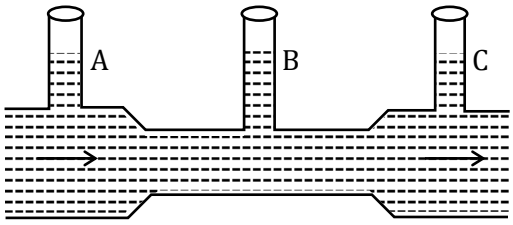
2. A body floats in a liquid contained in a beaker. The whole system as shown in figure falls freely under gravity. The upthrust on the body due to the liquid is :



(1) zero
(2) Equal to the weight of the liquid displaced by the body
(3) Equal to the weight of the body in air
(4) Equal to the weight of the immersed portion of the body in the liquid

PPF280

3. Three tubes A, B and C are connected to a horizontal pipe in which an ideal liquid is flowing. The radii of the tubes A, B and C at the junction are respectively 2 cm, 1cm and 2 cm. It can be said: -



(1) The height of the liquid in the tube A is maximum
(2) Height of the liquid in the tubes A and B is same
(3) Height of the liquid in the tubes A, B and C is same
(4) Height of the liquid in the A and C is same

PPF281

4. Two spherical metal balls (of the same material) with radii 0.2 cm and 0.3 cm are falling through a fluid at constant speeds. If the speed of the first ball (radius = 0.2 cm) is 4 cm/s. What is the speed of the other ball?

(1) 6 cm/s (2) 8 cm/s
(3) 9 cm/s (4) 12 cm/s

PPF282

5. Due to capillary action water rises to a height of 16.3 cm in a capillary of length 18 cm above the water level. If the tube is cut a t height of 12 cm: -

(1) Water will come as fountain from the capillary tube.
(2) Water will stay at a height of 12 cm in the capillary tube but radius of curvature at top will increase.
(3) Water will stay at a height of 12 cm in capillary tube but radius of curvature at top will decrease.
(4) Water will flow down the side of the capillary tube.

PPF283

6. Two identical air bubbles are rising in water. If they coalesce under isothermal condition to form a single air bubble, the combined air bubble would have -

(1) Higher excess pressure but smaller terminal velocity
(2) Higher excess pressure and higher terminal velocity
(3) Lower excess pressure and smaller terminal velocity
(4) Lower excess pressure but higher terminal velocity

PPF284

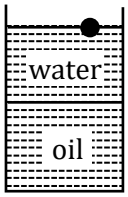
7. We have three beakers A, B and C containing three different liquids. They are stirred vigorously and placed on a table. Then, liquid which is -

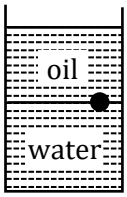
(1) Most viscous comes to rest at the earliest
(2) Most viscous comes to rest at the last
(3) Most viscous slow down earliest but comes to rest at the last
(4) All of them come to rest at the same time

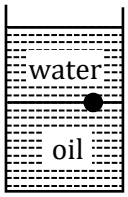
PPF285

8. A ball is made of a material of density ρ where $\rho_{water} > \rho > \rho_{oil}$ with ρ_{oil} and ρ_{water} representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position?

(1) 

(2) 

(3) 

(4) 

PPF286

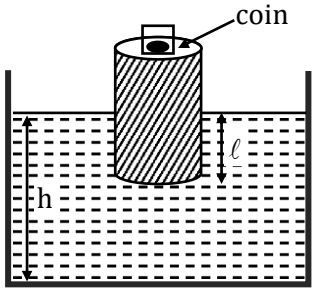
9. If diameter of a copper wire is 2 cm. Then find the required force to change the length of wire by 0.02% ($Y = 2 \times 10^{10}$ N/m^2) :-

(1) $16 \times 10^2 \pi$ N (2) $8 \times 10^2 \pi$ N

(3) $4 \times 10^2 \pi$ N (4) $1.6 \times 10^2 \pi$ N

PPF287

10. A wooden block, with a coin placed on its top, floats in water as shown in fig. the distance ℓ and h are shown. After some time the coin falls into the water. Then:



(1) ℓ decreases and h increases

(2) ℓ increases and h decreases

(3) Both ℓ and h increase

(4) Both ℓ and h decrease

PPF288

11. The surface energy of a liquid drop is E. It is sprayed into 1000 equal droplets. Then, its surface energy becomes:

(1) E (2) 10E (3) 100E (4) 1000E

PPF289

12. Calculate the approximate change in density of water in a lake at a depth of 400 m below the surface. The density of water at the surface is 1030 kg/m^3 and compressibility of water is 0.5×10^{-9} kg^{-1} /m^3

(1) 2 kg/m^3 (2) 5 kg/m^3

(3) 0.5 kg/m^3 (4) 3.2 kg/m^3

PPF290

13. The reading of spring balance when a block is suspended from it in air, is 60N. This reading is changed to 40 N when the block is immersed in water. The specific density of the block is

(1) 3 (2) $\frac{2}{3}$ (3) 6 (4) $\frac{3}{2}$

PPF291

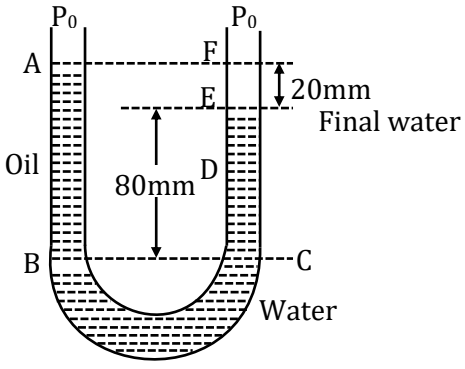
14. Water is flowing with velocity 4m/s in a cylinder of diameter 8cm, it is connected to a pipe with it end tip of diameter 2cm, calculate the velocity of water at this free end :-

(1) 4 m/s (2) 8 m/s

(3) 32 m/s (4) 64 m/s

PPF292

15. A U tube with both ends open to the atmosphere, is partially filled with water. Oil, which is immiscible with water, is poured into one side until it stands at a distance of 20 mm above the water level on the other side. Meanwhile the water rises by 40 mm from its original level. The density of the oil is :-



(1) 800 kg/m^3 (2) 400 kg/m^3

(3) 100 kg/m^3 (4) 650 kg/m^3

PPF293

16. Two equal drops are falling through air with a steady velocity of 5 cm/second. If two drops coalesce to form one drop then new terminal velocity will be–

(1) $5 \times (4)^{1/3}$ cm/s (2) $5\sqrt{2}$ cm/s

(3) $\frac{5}{\sqrt{2}}$ cm/s (4) 10 cm/s

PPF294

17. Equal masses of water and a liquid of relative density two are mixed together; then mixture has a relative density of :

(1) 2/3 (2) 4/3 (3) 3/2 (4) 3

PPF295

18. When an air bubble comes from the bottom of a lake to the top surface, then its radius:

(1) Increases (2) Decreases

(3) Does not change (4) Becomes zero

PPF296

19. A solid is completely immersed in an ideal liquid. The force exerted by the liquid on the solid will

(1) increase, if it is pushed deeper inside the liquid

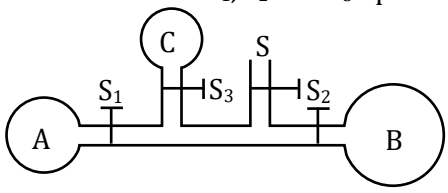
(2) change, if its orientation is changed keeping it inside the liquid

(3) decrease, if it is taken partially out of the liquid

(4) All of these

PPF297

20. The given figure shows three soap bubbles A, B and C are prepared by blowing the capillary tube with stop cocks S, S_1, S_2 and S_3. With stop cock S closed and S_1, S_2 and S_3 opened, then-



(1) C and A will both start collapsing with volume of B increasing

(2) B will start collapsing with volumes of A and C increasing

(3) C will start collapsing with volumes of A and B increasing

(4) volumes of A, B and C will become equal at equilibrium

PPF298

21. A volume V of a viscous liquid flows per unit time due to pressure head of ΔP along a pipe of diameter D and length l. Instead of this pipe a set of four pipes each of diameter D/2 and length $2l$ is connected in parallel to the same pressure head of ΔP. Now the volume of the liquid flowing per unit time is -

(1) V (2) V/4 (3) V/6 (4) V/8

PPF299

22. A space 2.5 mm wide between two large plane surfaces is filled with oil. Force required to drag a very thin plate of area of one side 0.5 m^2 just midway of the surfaces at a velocity of 0.5 m/sec is 1 N. The coefficient of viscosity in kgs/m^2 is -

(1) 1×10^{-2} (2) 2.5×10^{-2}

(3) 5×10^{-2} (4) 2.5×10^{-3}

PPF300

23. Pressure at a point inside a liquid at rests does not depend on -

(1) The depth of the point below the surface of the liquid

(2) The nature of the liquid

(3) The acceleration due to gravity at that point

(4) Total weight of the liquid in the beaker

PPF301

24. For an aerofoil which of the following statement(s) is/are correct -

(I) The orientation of the wings relative to flow direction causes the streamline

(II) Flow speed on top is lower than that below it

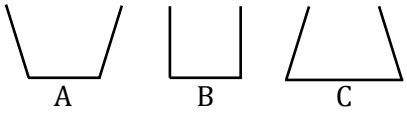
(III) Upward force resulting in dynamic lift of the wings

(1) Only I (2) Only III

(3) Only I and II (4) Only I and III

PPF302

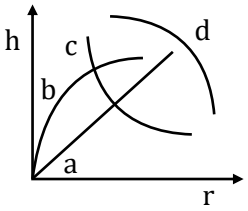
25. The three vessel shown in figure have same base area. Equal volumes of a liquid are poured in the three vessel. The force on the base will be-



(1) Maximum in vessel A
(2) Maximum in vessel B
(3) Maximum in vessel C
(4) Equal in all the vessels

PPF303

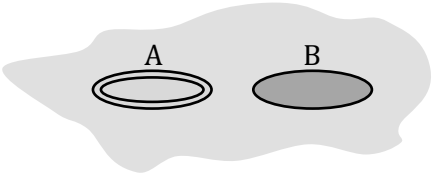
26. Which of the following graphs may represent the relation between the capillary rise (h) and the radius (r) of the capillary?



(1) a (2) b (3) c (4) d

PPF304

27. A rigid ring A and a rigid thin disk B both made of same material, when gently placed on water, just manage to float due to surface tension as shown in the figure. Both the ring and the disk have same radius. What can you conclude about their masses (thickness may be different)?



(1) Both have the same mass.
(2) Mass of the ring is half of that of the disk.
(3) Mass of the ring is double to that of the disk.
(4) More information is needed to decide.

PPF305

28. The graph between terminal velocity (along y-axis) and square of radius (along x-axis) of spherical body of density ρ allowed to fall through a fluid of density σ is a :-
(1) Straight line with positive slope
(2) Straight line with negative slope
(3) Straight line perpendicular to x-axis
(4) Straight line perpendicular to y-axis

PPF306

29. The young's modulus and shear modulus are relevant
(1) for liquids and solids both
(2) only for solids
(3) only for liquids
(4) neither solids nor liquids

PPF307

30. Which of the following has highest compressibility: -
(1) Solid (2) Liquids
(3) Gases
(4) Depend on material

PPF308

31. Pressure is a .....Quantity.
(1) Vector (2) Scalar
(3) Tensor (4) (1) and (2) both

PPF309

32. Swinging of a spinning ball mid-air is explained in ..... .
(1) Torricelli's law (2) Relative density
(3) Magnus effect (4) None of the above

PPF310

33. Dimensional formula of specific gravity is
(1) $[M^0LT^{-2}]$ (2) $[M^0L^0T^0]$
(3) $[M^1L^1T^{-2}]$ (4) $[M^1L^0T^{-2}]$

PPF311

34. Equation of continuity is based on
(1) Conservation of mass
(2) Conservation of energy
(3) Conservation of charge
(4) None of the above

PPF312

35. Hydraulic brakes and Hydraulic lifts are based on
(1) Torricelli's law (2) Magnus effect
(3) Pascal's law (4) None of the above

PPF313

36. If rise in capillary is greater than the height of capillary itself then
(1) liquid will come out of tube slowly
(2) liquid comes out of tube in the form of small fountain
(3) liquid just reaches to the top and adjust its radius of curved surface
(4) Can't say

PPF314

37. The correct conversion is
(1) 1 atm = 1.01×10^5Pa
(2) 1 torr = 133 Pa
(3) 1 mm of Hg = 1 torr
(4) all of the above

PPF315

38. Base of a dam is thicker as compared to the top, Reason for this is.....
(1) To make dam look beautiful
(2) To make it higher
(3) Pressure increases with depth so greater force is applicable on same area at the bottom as compared to top.
(4) None of the above

PPF316

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	4	3	2	4	1	3	3	4	2	1	1	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	1	3	1	4	4	4	4	3	3	3	1	2	3
Question	31	32	33	34	35	36	37	38							
Answer	2	3	2	1	3	3	4	3							

Thermal Physics

1. Which of the following relation is correct for temperature 50° F, 50° C, 50 K

(1) 50° F > 50° C > 50 K

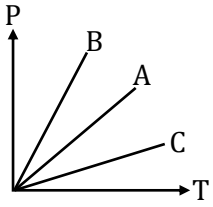
(2) 50° F = 50° C = 50 K

(3) 50° C > 50° F > 50 K

(4) 50° C > 50 K > 50° C

PTM532

2. Pressure versus temperature graph of an ideal gas at constant volume is shown by a straight line-A in figure. If the mass of the gas is doubled and volume is halved, the corresponding pressure versus temperature graph is given by line :

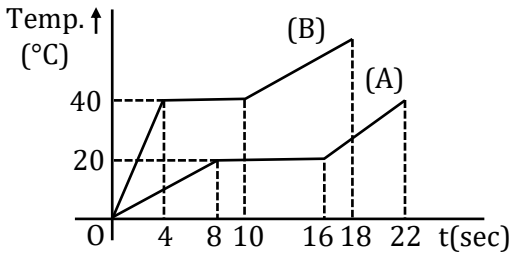


(1) A (2) B

(3) C (4) None

PTM533

3. Two solid objects A and B of equal mass are heated at a uniform rate. In the given figure graphs A and B respectively represent their temperature variation. What is the ratio of (i) latent heats of A and B (ii) Specific heats of A and B in solid state.



(1) $\frac{4}{3},1$ (2) $\frac{4}{3},4$

(3) $\frac{3}{4},\frac{1}{4}$ (4) $\frac{3}{4},1$

PTM534

4. Consider the following two statements

(A) If heat is added to a system, its temperature must increase.

(B) If positive work is done by a system in a thermodynamic process, its volume must increase.

(1) Both A & B are correct

(2) A is correct but B is wrong

(3) B is correct but A is wrong

(4) Both A & B are wrong

PTM535

5. A Vessel contain 2 non-reactive gases Neon and Oxygen. The ratio of their partial pressure is 3 : 2 find ratio of No. of molecules.

(1) 2:3 (2) 3:2 (3) 1:1 (4) 3:5

PTM536

6. Newton's law of cooling is a special case of

(1) Wien's displacement law

(2) Kirchoff's law

(3) Stefan's law

(4) Plank's law

PTM537

7. Two gases are separated by fixed diathermic wall with initial state (P_A V_A) and (P_B V_B). Choose correct statement-

(1) Thermodynamics state of A and B will not change with time

(2) The state of two system will change until the equilibrium is attained

(3) Both (1) and (2) possible

(4) None

PTM538

8. At a certain temperature (T_1) most probable wavelength in the radiation emitted by blackbody is 12000 Å. On increasing the temperature to T_2, total radiation emitted is increased to 256 times. Most probable wavelength now becomes.

(1) 6000 Å (2) 9000 Å

(3) 15000 Å (4) 3000 Å

PTM539

9. If work is done on the system in an adiabatic process
(1) $T_f < T_i$
(2) $T_f = T_i$
(3) $T_f > T_i$
(4) Information insufficient

PTM540

10. Choose the correct option.
(1) $C_P - C_V = R$ is true for monoatomic ideal gas
(2) $C_P - C_V = R$ is true for poly atomic ideal gas
(3) $C_P - C_V = R$ is true for any type of ideal gas
(4) None of the above

PTM541

11. A composite wall of area A is made of equal thickness of lead and iron having thermal conductivities K and 2K respectively. The temperatures on the two sides of the composite wall are 100°C & 0°C with lead layer on the hotter side. Calculate the steady state temperature on the lead - iron interface & also the equivalent thermal conductivity:
(1) $\left(\frac{100}{3}\right)$°C, $\frac{4K}{3}$ (2) 50°C, 3K
(3) $\left(\frac{100}{3}\right)$°C, $\frac{3K}{2}$ (4) 50°C, $\frac{4K}{2}$

PTM542

12. Which of the following statements is true?
(A) With increase in pressure boiling point of water increases
(B) Heat cannot be stored in a body
(C) With increase in pressure melting point of ice decreases
(D) Vapours can be directly converted into solid
(1) Only (A), (C), (D)
(2) All (A), (B), (C), (D)
(3) Only (A), (C)
(4) Only (C), (D)

PTM543

13. Match the following column

	Column-I		Column-II
(a)	PV=KNT	(1)	Gay Lussac's law
(b)	$P \propto \frac{1}{v}$	(2)	Boyle's law
(c)	$P \propto T$	(3)	Ideal Gas Equation
(d)	$v \propto T$	(4)	Charle's law

	a	b	c	d
(1)	3	1	2	4
(2)	3	2	4	1
(3)	2	3	1	4
(4)	3	2	1	4

PTM544

14. If the atmospheric temperature is below 0°C and ice begins to form at t = 0, then the thickness of ice formed upto a time 't' will be directly proportional to
(1) t^4 (2) t^2 (3) t (4) $t^{1/2}$

PTM545

15. We took two separate gas with same molecular density. If ratio of diameter of their molecule is 4 : 1 then ratio of their mean free path is
(1) 1 : 4 (2) 4 : 1 (3) 2 : 1 (4) 1 : 16

PTM546

16. Cooking is difficult on hills because
(1) atmospheric pressure is higher
(2) atmospheric pressure is lower
(3) boiling point of water is reduced
(4) Both (2) and (3)

PTM547

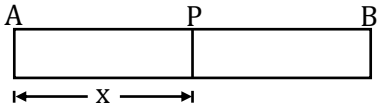
17. A pan filled with hot food cools from 50°C to 49.9°C in 5sec. How long will it take to cool from 40°C to 39.9°C, if room temperature is 30°C :
(1) 2.5 sec (2) 10 sec
(3) 20 sec (4) 5 sec

PTM548

18. A glass of water is stirred and then allowed to stand until the water stops moving. Then internal energy of water -
(1) remains constant
(2) will increase
(3) will decrease
(4) nothing can be said

PTM549

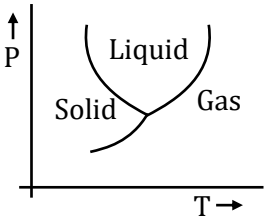
19. Heat is flowing steadily from A to B. Temperature T at point P, at distance x from A is such that :-



(1) T decreases linearly with x
(2) T increases linearly with x
(3) T decreases exponentially with x
(4) T increases with x as $T \propto x^2$

PTM550

20. For the phase diagram of water given in figure, curves OA, AB and AC are respectively.



(1) Sublimation curve, vaporisation curve and fusion curve
(2) Sublimation curve, fusion curve and vaporisation curve
(3) Fusion curve, vaporisation curve and sublimation curve
(4) Fusion curve, sublimation curve and vaporisation curve

PTM551

21. Which of the following statements is/are correct?
(I) Convection is a mode of heat transfer by actual motion of matter.
(II) Convection is possible only in gases.
(III) Convection can be natural or forced.
(1) Only I (2) Only II
(3) Both I and III (4) None of these

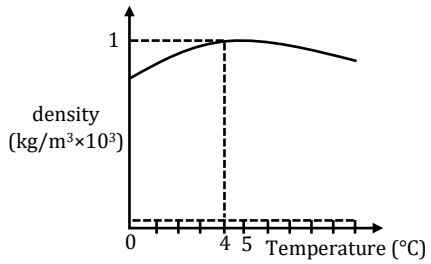
PTM552

22. Find out the increase in moment of inertia I of a uniform rod (Coefficient of linear expansion α) about its perpendicular bisector when its temperature is slightly increased by ΔT.
(1) $2I\alpha\Delta T$ (2) $4I\alpha\Delta T$
(3) $6I\alpha\Delta T$ (4) $3I\alpha T$

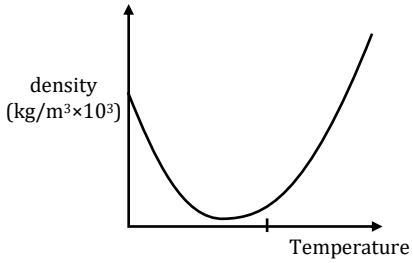
PTM553

23. Which of the following graph shows the variation of density of water with increase in temperature.

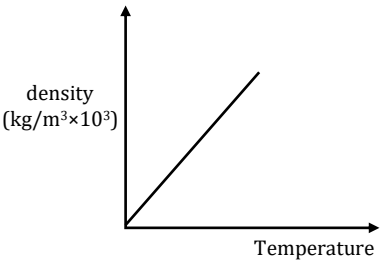
(1)



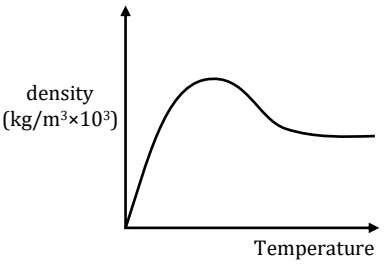
(2)



(3)



(4)



PTM554

24. Water is used as coolant in automobiles radiators because.
(1) It has low density
(2) It has low specific heat
(3) It has high specific heat
(4) It has high density

PTM555

25. A sphere, cube and a thin plate, all of same material and mass, are initially heated to same high temperature. Then
(1) Plate will cool fastest and cube the slowest
(2) Sphere will cool fastest and cube the slowest
(3) Plate will cool fastest and sphere the slowest
(4) cube will cool fastest and plate the slowest

PTM556

26. Which of the following is correct according to Claussius statement of second law of thermodynamics.
(1) It is possible to transfer heat from a low temp. reservoir to a high temp. reservoir.
(2) No heat engine can have 100% efficiency.
(3) No process is possible whose sole result is the transfer of heat from a colder object to a hotter object
(4) None of these

PTM557

27. Which of the following statement is correct.
(1) If heat is added to a system, its temperature must increase.
(2) If positive work is done by a system in a thermodynamic process, its volume must increase.
(3) In free expansion, heat is necessarily absorbed by the system.
(4) In a process, the initial pressure and volume are equal to the final pressure and volume then the net work done by the system in the process must be zero.

PTM558

28. Hydrogen and nitrogen are at same temperature. The molecules of which one of them have more kinetic energy: -
(1) Hydrogen
(2) Nitrogen
(3) Both have equal amount of energy
(4) Depends upon actual value of temperature

PTM559

29. Pressure of an ideal gas is increased by keeping temperature constant. What is its effect on kinetic energy of molecules.
(1) Increases
(2) Decreases
(3) No change
(4) Cannot be determined

PTM560

30. The root mean square speed of the ideal gas molecule is doubled if.
(1) Pressure is doubled at constant volume
(2) Pressure is made four times at constant volume
(3) Volume is doubled at constant pressure
(4) Volume is increased by about 42% at constant pressure

PTM561

31. For an ideal gas, coefficient of volume expansion at constant pressure is equal to: -
(1) T
(2) $\frac{1}{T}$
(3) T^2
(4) Not depends on temp.

PTM562

32. A bimetallic strip is made of aluminium and steel ($\alpha_{al} > \alpha_{steel}$). On heating the strip will: -
(1) remain straight
(2) get twisted
(3) will bend with aluminium on concave side
(4) will bend with steel on concave side

PTM563

33. As temperature is increased, the time period of pendulum.
(1) increases as its length increases even though its centre of mass still remains at centre of the bob
(2) decreases as its length increases even though its centre of mass still remains at centre of the bob
(3) increases as its length increases due to shifting of centre of mass below centre of the bob
(4) decreases as its effective length remains same but centre of mass shift above centre of bob

PTM564

34. Two vessel of the same volume contain the same gas at same temperature. If pressure in the vessel are in ratio of 1 : 2 then-
(1) Ratio of v_{rms} is 1 : 2
(2) Ratio of kinetic energy is 1 : 1
(3) Ratio of average velocity is 1 : 2
(4) Ratio of number of molecules is 1 : 2

PTM565

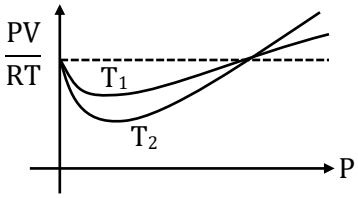
35. One calorie is defined to be the amount of heat required to raise the temperature of 1g of water from ..... at 1 atm.
(1) 0°C to 1°C (2) 10°C to 11°C
(3) 4°C to 5°C (4) 14.5°C to 15.5°C

PTM566

36. Under isobaric condition if the temperature of a room increases then-
(1) Total KE of the molecules increases.
(2) Total KE of the molecules decreases.
(3) Total KE of the molecules remain same.
(4) The density of air increases.

PTM567

37. In given figure dotted line indicates: -



(1) Real gas (2) Ideal gas
(3) Both (4) None

PTM568

38. In given figure what is relation between T_1 & T_2
(1) $T_1 < T_2$ (2) $T_1 > T_2$
(3) $T_1 = T_2$ (4) None

PTM569

39. Which one is an extensive variable: -
(1) Pressure (2) Temperature
(3) Density (4) Volume

PTM570

40. Which one is an intensive variable: -
(1) Internal energy (2) Volume
(3) Mass (4) Temperature

PTM571

41. In given option. Choose intensive variable: -
(1) ΔQ (2) ΔU
(3) $P\Delta V$ (4) None

PTM572

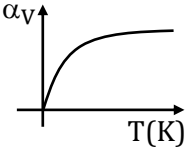
42. In P-T diagram of CO_2. Triple point exist: -
(1) 27°C (2) 0.01°C
(3) –56.6°C (4) –78.5°C

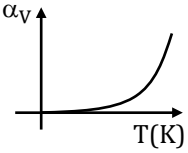
PTM573

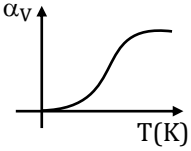
43. In P-T diagram of CO_2, Triple point exist at pressure (in atm): -
(1) 1 atm (2) 5.11 atm
(3) 6.11 atm (4) 56 atm

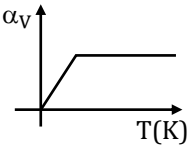
PTM574

44. Which of the following graphs correctly shows variation of coefficient of volume expansion of copper with temperature.

(1) 

(2) 

(3) 

(4) 

PTM575

45. When thermometer is taken from the melting ice to a warm liquid the mercury level rises to $\left(\frac{2}{5}\right)^{th}$ of the distance between the lower and upper fixed points. The temperature of liquid in kelvin is: -
(1) 217.15 (2) 313.15
(3) 220 (4) 330

PTM576

46. Heat capacity of substance is infinite. It means: -
(1) heat is given out
(2) heat is taken in
(3) No change in temperature whether heat is taken in or given out
(4) All of these

PTM577

47. Match the following: -

Column-I		Column-II	
(A)	Conversion of liquid into solid	(P)	Regelation
(B)	Conversion of liquid into vapour	(Q)	Sublimation
(C)	Conversion of solid into vapour directly	(R)	Fusion
(D)	Melting of ice caused by pressure	(S)	Vaporisation

(1) A → r ; B → q ; C → p ; D → s
(2) A → r ; B → s ; C → q ; D → p
(3) A → q ; B → p ; C → s ; D → r
(4) A → p ; B → q ; C → r ; D → s

PTM578

48. The equatorial and polar regions of the earth receives unequal solar heat. The convection current arising due to this is called: -
(1) Land breeze
(2) Sea breeze
(3) Trade wind
(4) Tornado

PTM579

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49. A uniform metallic rod rotates about its perpendicular bisector with constant Angular speed. If it is heated uniformly to raise its temperature slightly.

(1) Its speed of rotation increases

(2) Its speed of rotation decrease.

(3) Its speed of rotation remain same

(4) Its speed increases because its moment of inertia increase.

PTM580

50. Heat is associated with: -

(1) K.E. of random motion of molecules

(2) K.E. of orderly motion of molecules

(3) Total K.E. of random and orderly motion of molecules

(4) K.E. of random motion in some cases and K.E. orderly motion in other

PTM581

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	2	3	2	3	2	4	3	3	1	2	4	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	2	1	2	3	1	1	3	3	3	2	3	3	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	4	1	4	4	3	2	2	4	4	4	3	2	3	2
Question	46	47	48	49	50										
Answer	3	2	3	2	1										

Oscillations

1. General vibration of a polyatomic molecule about its equilibrium position is :-
(1) Simple harmonic motion
(2) Neither periodic nor SHM
(3) Superposition of SHMs
(4) None

POS242

2. The equation of motion of a particle is $x = a\cos(\alpha t)^2$ the motion is :-
(1) Periodic but not oscillatory
(2) Periodic and oscillatory
(3) Oscillatory but not periodic
(4) Neither periodic nor oscillatory

POS243

3. A body is doing SHM, which is not true: -
(1) Average total energy per cycle is equal to its maximum kinetic energy
(2) Average kinetic energy per cycle is equal to half of its maximum kinetic energy
(3) Mean velocity over a complete cycle is equal to $\frac{2}{\pi}$ times of its max. velocity
(4) Root mean square velocity is $\frac{1}{\sqrt{2}}$ times of its maximum velocity

POS244

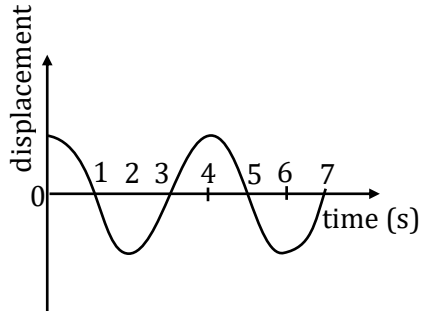
4. In SHM phase difference between displacement and velocity is: -
(1) Zero (2) π
(3) $\pi/2$ (4) None of these

POS245

5. A displacement equation of a particle executes S.H.M is $x = \sin \omega t$, then :-
(a) Mean position x = 0
(b) Mean position $x = \frac{1}{2}$
(c) Time period $= \frac{\pi}{\omega}$ s
(d) Time period $= \frac{2\pi}{\omega}$
(1) a, d (2) b,c (3) a,c (4) b,d

POS246

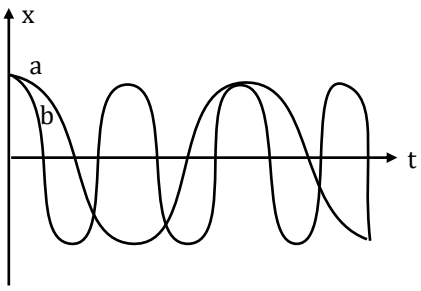
6. Displacement vs time curve for a particle executing S.H.M. is shown in fig given below. Choose the correct statement:



(1) Phase of the oscillation is same at t = 0s and t = 2s
(2) Phase of the oscillation is same at t = 2s and t = 6s
(3) Phase of the oscillation is same at t = 1s and t = 7s
(4) Phase of the oscillation is same at t = 1s and t = 3 s

POS247

7. Choose incorrect option : -



(1) Amplitude of both are same
(2) Both curve have same frequency
(3) Initial phase of both curve are same
(4) Time period of a is twice of time period of b

POS248

8. When an automobile of mass 4000kg is placed on its suspension system then it sags by 20 cm. Force constant of each shock absorber of this 4-wheeler, is
(1) 2×10^5 Nm^{-1} (2) 4×10^5 Nm^{-1}
(3) 5×10^4 Nm^{-1} (4) 2×10^4 Nm^{-1}

POS249

9. For small linear oscillations of a simple pendulum of having a bob of mass m, its force-constant k is proportional to the mass as: -

(1) m (2) $m^{1/2}$

(3) $m^{-1/2}$ (4) m^{-1}

POS250

10. For large angles, the oscillations of a simple pendulum are not SHM then their time period is:-

(1) $T = 2\pi\sqrt{\frac{\ell}{g}}$ (2) $T > 2\pi\sqrt{\frac{\ell}{g}}$

(3) $T < 2\pi\sqrt{\frac{\ell}{g}}$ (4) $T << 2\pi\sqrt{\frac{\ell}{g}}$

POS251

11. Acceleration due to gravity on the surface of moon is 1.63 m/s² then find length of a second's pendulum on the moon's surface.

(1) 6m (2) 0.167 m

(3) 1m (4) 0.042m

POS252

12. Equation of motion of a particle executing SHM is represented as $x = -2\sin\left(3t+\frac{\pi}{3}\right)$.

It can also be represented as: -

(1) $x = 2\sin\left(3t+\frac{\pi}{3}\right)$

(2) $x = 2\sin\left(3t+\frac{4\pi}{3}\right)$

(3) $x = 2\cos\left(3t+\frac{\pi}{3}\right)$

(4) $x = 2\cos\left(3t+\frac{4\pi}{3}\right)$

POS253

13. Equation of motion of a particle executing SHM is represented as, $x = -3\cos\left(t-\frac{\pi}{6}\right)$.

It can not be represented as : -

(1) $x = -3\sin\left(t+\frac{\pi}{3}\right)$ (2) $x = 3\cos\left(t+\frac{5\pi}{6}\right)$

(3) $x = -3\sin\left(t+\frac{\pi}{6}\right)$ (4) $x = 3\sin\left(t+\frac{4\pi}{3}\right)$

POS254

14. Equation of motion of four SHM are given below: -

(A) $x = 2\sin\left(3t+\frac{\pi}{2}\right)$ (B) $x = 3\sin\left(3t+\frac{\pi}{3}\right)$

(C) $x = 3\sin\left(2t+\frac{\pi}{6}\right)$ (D) $x = 2\sin\left(2t+\frac{\pi}{4}\right)$

The two equations which are different in both amplitude and frequency, are: -

(1) B and C (2) B and D

(3) A and B (4) C and D

POS255

15. Equation of motion of a particle is represented by $x = 2\sin\left(4t+\frac{\pi}{4}\right)$. It can also be represented as

(1) $x = 2\sin\left(4t-\frac{\pi}{4}\right)$

(2) $x = 2\sin\left(4t+\frac{3\pi}{4}\right)$

(3) $x = 2\cos\left(4t+\frac{7\pi}{4}\right)$

(4) $x = 2\cos\left(4t-\frac{7\pi}{4}\right)$

POS256

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	3	3	2	2	2	3	1	2	2	2	3	2	3

Wave Motion

1. The time taken by a particle in reaching from through to crest in a transverse wave is: -

(1) $\frac{T}{4}$ (2) $\frac{T}{2}$ (3) $\frac{3T}{4}$ (4) T

PWA276

2. Calculate the velocity of transverse waves in a copper wire 1 mm^2 in cross - section under the tension produced by 1 kg wt. The linear mass density of copper is = 8.93×10^{-3} kg/m.

(1) 22.12 m/s (2) 33.12 m/s
(3) 44.12 m/s (4) 55.12 m/s

PWA277

3. Speed of sound in air at room temperature is approximately 343 m/s. What can be the speed of sound in Hydrogen at room temperature approximately.

(1) 1284 m/s (2) 2089 m/s
(3) 686 m/s (4) 440 m/s

PWA278

4. An air column of length L closed at one end is oscillating in 2 overtone. Pressure change in air is minimum from closed end at distance.

(1) $\frac{L}{3}$ (2) $\frac{L}{5}$ (3) $\frac{L}{7}$ (4) Zero

PWA279

5. The extension in string obeying hook's law is x. The speed of wave in the stretched string v. If the extension in string is increased to 1.5 x, the speed of wave will be

(1) 1.22 v (2) 0.61 v
(3) 1.50 v (4) 0.75 v

PWA280

6. Two sitar strings A and B playing the note 'Dha' are slightly out of tune and produces beats of frequency 5Hz. The tension of the string B is slightly increased and the beat frequency is found to decrease to 3Hz. What is the original frequency of B if the frequency of A is 427 Hz?

(1) 422 Hz (2) 432 Hz
(3) 424 Hz (4) 430 Hz

PWA281

7. What should be the length of a closed organ pipe to produce a fundamental frequency of 512 Hz at 0°C

(1) 13.2 cm (2) 16.2 cm
(3) 14.2 cm (4) 15.2 cm

PWA282

8. A wave travelling along a string is described by

$y(x, t) = 0.005 \sin(20\pi t - 10\pi x + \frac{\pi}{2})$ cm

where x is in cm and t is in sec. Calculate the displacement y of the wave at a distance x = 1 cm and time t = 1 sec.?

(1) y = 0.0025 cm (2) y = – 0.0025 cm
(3) y = 0.005 cm (4) y = – 0.005 cm

PWA283

9. In open organ pipe, the position of antinode are obtained at :-

(1) $0, \frac{\lambda}{2}, 2\lambda$ (2) $0, \frac{\lambda}{2}, \lambda$
(3) $\frac{\lambda}{4}, \frac{3\lambda}{4}, \lambda$ (4) $\frac{5\lambda}{4}, \frac{3\lambda}{4}$

PWA284

10. If speed of sound at 0°C is 330 m/s then what is the speed of sound at 20°C.

(1) 322 m/s (2) 332 m/s
(3) 342 m/s (4) 352 m/s

PWA285

11. The speed of sound according to Laplace corrections :-

(1) $\sqrt{\frac{p}{\gamma\rho}}$ (2) $\sqrt{\frac{p\rho}{\gamma}}$
(3) $\sqrt{\frac{\gamma p}{\rho}}$ (4) $\sqrt{\frac{\gamma\rho}{p}}$

PWA286

12. The length of an organ pipe open at both ends is 0.5 m. Calculate the fundamental frequency of the pipe, if velocity of sound in air be 350 m/s. If one end of the pipe is closed, then the fundamental frequency will be :-

(1) 350, 700 (2) 700, 350
(3) 175, 350 (4) 350, 175

PWA287

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13. The equation of a progressive wave moving in +ve direction is given by

(1) $y = A\sin\left[2\pi\left(\frac{Vt}{\lambda} - x\right)\right]$

(2) $y = A\sin\frac{2\pi}{\lambda}\left(Vt - \frac{x}{\lambda}\right)$

(3) $y = A\sin\frac{2\pi}{\lambda}(Vt - x)$

(4) $y = A\sin\frac{2\pi}{\lambda}(x + Vt)$

PWA288

14. Which relation is correct for speed of sound wave in solid (s), liquid (ℓ) & Gases (g) :-

(1) $V_s > V_\ell > V_g$ (2) $V_s < V_\ell < V_g$

(3) $V_s = V_\ell = V_g$ (4) $V_\ell > V_g > V_s$

PWA289

15. A pipe, 30 cm long, is open at both ends. which harmonic mode of the pipe resonates a 1.1 kHz source? will resonance with the same source be observed if one end of the pipe is closed?

Take the speed of sound in air as 330 m/s.

(1) 4th Harmonic, yes (2) II Harmonic, yes

(3) 4th Harmonic, No (4) II Harmonic, No

PWA290

16. The displacement of the wave given by equation $y(x, t) = a\sin(kx - \omega t + \phi)$ where $\phi = 0$ at point x and t = 0 is same as that at point:-

(1) $x + 2n\pi$ (2) $x + \frac{2n\pi}{k}$

(3) $kx + 2n\pi$ (4) None

PWA291

17. The transverse stationary wave on a stretched wire (Clamped at its both ends) is given by –

$$y(x, t) = 0.06\sin\left(\frac{2\pi}{3}x\right)\cos(120\pi t)$$

where x and y are in m and t in second. The length of string is 1.5 m and it's mass is 3×10^{-2}kg then all the points on the wire vibrate with -

(1) Same phase (2) Same amplitude

(3) Same energy (4) Different frequency

PWA292

18. As per laplace correction, the compression and expansion of gas for longitudinal wave are

(1) Isothermal (2) Adiabatic

(3) Isobaric (4) Isochoric

PWA293

19. Phase difference between two points on both sides of a node is

(1) same phase (2) π

(3) $\frac{\pi}{2}$ (4) none of the above

PWA294

20. Two particles lies somewhere between two consecutive nodes has

(1) same amplitude

(2) same phase

(3) same oscillating frequency

(4) (2) and (3) both

PWA295

21. Dimensional formula of $\sqrt{\frac{T}{\mu}}$ is .....

(where $T \rightarrow$ tension in string

$\mu \rightarrow$ linear mass density of string)

(1) $[M^0L^0T^{-1}]$ (2) $[M^0L^1T^{-1}]$

(3) $[M^1L^2T^{-1}]$ (4) $[M^0L^{-1}T]$

PWA296

22. A progressive wave is represented by

$Y = 5\sin(50\pi t - 4\pi x)$

where x and y are in cm and t is in second. Its frequency and wavelength are

(1) 25 Hz, 4π (2) 25 Hz, 0.5m

(3) 50πHz, 4πm (4) 50πHz, 0.5m

PWA297

23. For beats, two waves super positioning must have: -

(1) same frequencies

(2) different frequencies

(3) same amplitude

(4) different amplitudes

PWA298

24. For any longitudinal wave propagating in a medium the property of medium needed is

(1) Shear modulus

(2) Bulk modulus

(3) Both shear and Bulk modulus

(4) Young's modulus

PWA299

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25. Dimensional formula of propagation constant is: -
(1) [L] (2) $[L^{-1}]$
(3) $[LT^{-1}]$ (4) $[T^{-1}]$

PWA300

26. In a progressive wave ..... transfer takes place
(1) particle
(2) speed
(3) energy
(4) none of the above

PWA301

27. When a wave-pulse super impose over its own reflection (from rigid surface), particle at the point remains at their mean position. For this point of superposition particles have
(1) kinetic and potential both energies
(2) only kinetic energy
(3) only potential energy
(4) no energy as particle one of their mean position

PWA302

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	1	2	1	1	2	3	2	3	3	4	3	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27			
Answer	2	1	2	2	4	2	2	2	2	2	3	2			